

SHAKURA-SUNYAEV ALPHA ACCRETION DISK

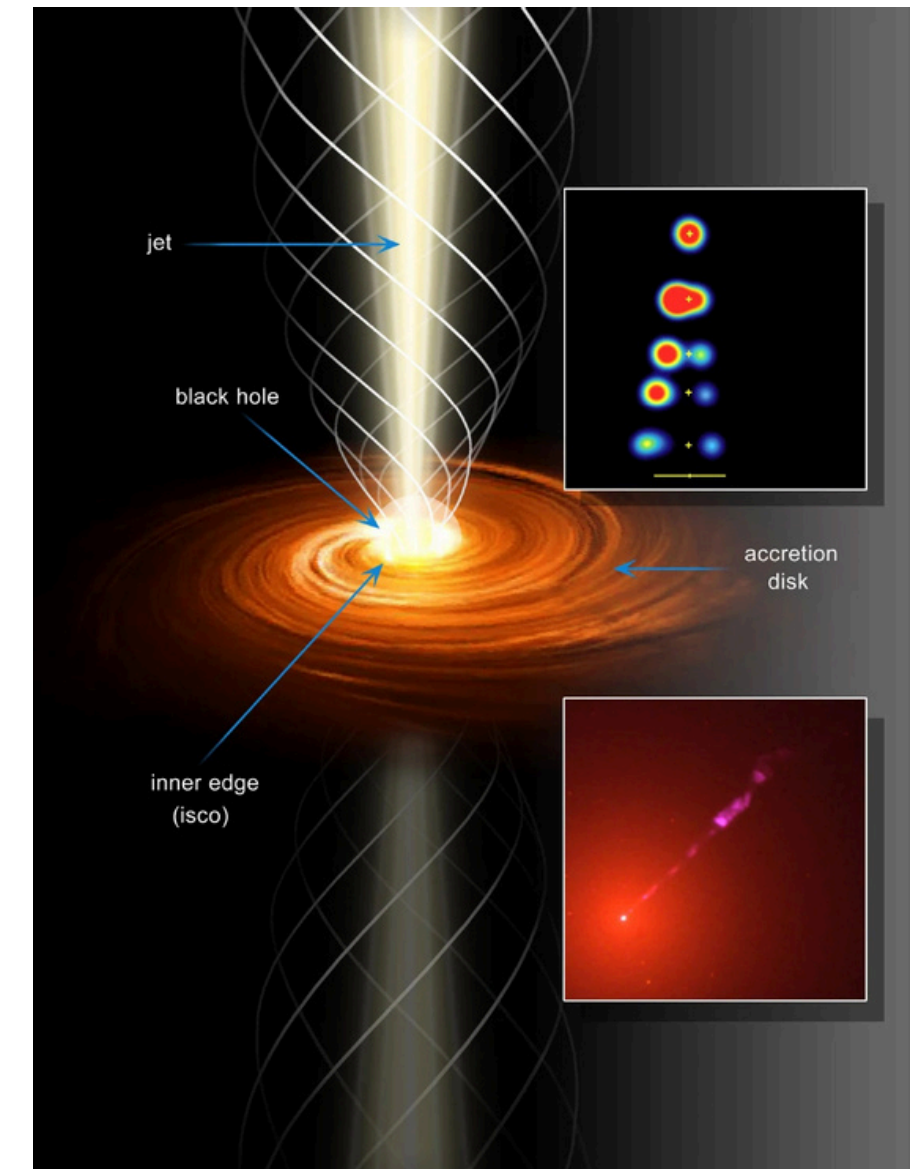
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Physical Problem

We consider a central Schwarzschild black hole with infalling matter that accretes onto it, the outward flow of angular momentum of the accreting matter leads to the formation of an accretion disk around the black hole.

We can describe the viscosity ν of the disk through an adimensional viscosity parameter $\alpha \rightarrow \nu = \alpha c_s H$

This formalism was introduced and studied by Shakura and Sunyaev in 1972 and so take the name of *Shakura-Sunyaev α accretion disk*



Assumptions:

- Stationary
- Axially Symmetric
- Geometrically Thin
- Optically Thick
- Radiation Efficient
- Alpha

Equations

$$\Sigma\nu = \frac{\dot{M}}{3\pi} f(R), \quad f(R) \equiv 1 - \left(\frac{R_*}{R} \right)^{1/2}, \quad Q^+ \equiv \frac{3GM\dot{M}}{4\pi R^3} f(R) = \frac{8\sigma T_c^4}{3\kappa\Sigma}$$

$$\nu = \alpha c_s H$$

$$H = \frac{c_s}{\Omega_K}, \quad \Omega_K = \sqrt{\frac{GM}{R^3}}, \quad c_s^2 = \frac{P_c}{\rho_c}, \quad \rho_c = \frac{\Sigma}{2H}$$

$$P_c = \frac{\rho_c k T_c}{\mu m_p} + \frac{a T_c^4}{3}, \quad \kappa = \kappa_{es} + \kappa_0 \rho_c T_c^{-7/2}$$

Goals

We want to solve the system of equations at fixed radius, mass, accretion rate and viscosity parameter for Σ , T_c , H , ρ_c

We then vary our parameters to see how our 4 variables change, in particular we're going to analyze:

a) Radial Profiles

$$\begin{array}{cccc}\Sigma(R) & T_c(R) & H(R)/R & \rho_c(R) \\ P_{gas}(R) & P_{rad}(R) & \beta(R) & \end{array}$$

Goals

b) Pressure Regimes and Analytical Limits

Identification of different regions of the disk with the change of dominant pressure or opacity

c) Peak Temperature Scaling

$$T_c^{MAX}(\dot{M}) \quad T_{eff}^{MAX}(\dot{M}) \quad T_{eff}^{MAX}(M)$$

d) Luminosity Profile

Numerical integration of the luminosity and analytical check

e) Emergent Spectrum

$$L(\nu)$$

Numerical Methods

To analyse the system I had to use 3 numerical algorithms:

- Newton-Rhapson
- Gauss-Jordan
- Simpson

Newton-Rhapson is a root-finding algorithm, while Gauss-Jordan is a linear system solver algorithm, I used them together to solve my non linear system of equations.

Simpson is an integration method that I used to integrate the total luminosity and the monochromatic luminosity.

Newton-Rhapson

For a non linear system of equations, the Newton-Rhapson algorithm is written as follows:

$$x_{k+1} = x_k - J^{-1}(x_k)F(x_k) \rightarrow J(x_k)\Delta x = -F(x_k)$$

Which means that I had to write my system of equations accordingly, compute the Jacobian and pick an initial guess.

$$\begin{cases} 3Q^+ \kappa_{es} \Sigma + 3Q^+ \Sigma \kappa_0 \rho_c T_c^{-7/2} - 8\sigma T_c^4 = 0 \\ 3\Sigma k T_c + 2a T_c^4 H \mu m_p - 3\mu m_p H^2 \Omega_K^2 \Sigma = 0 \\ 3\pi \Sigma \alpha \Omega_K H^2 - \dot{M} f(R) = 0 \\ 2\rho_c H - \Sigma = 0 \end{cases} \quad \begin{cases} T_c = \left(\frac{Q^+ \kappa_{es} \dot{M} f(R) \mu m_p \Omega_K}{8\pi \sigma \alpha k} \right)^{1/5} \\ H = \frac{1}{\Omega_K} \sqrt{\frac{k}{\mu m_p} T_c} \\ \Sigma = \frac{\dot{M} f(R)}{3\pi \alpha \Omega_K H^2} \\ \rho_c = \frac{\Sigma}{2H} \end{cases}$$

Newton-Rhapon

$$J = \begin{pmatrix} 3Q^+(\kappa_{es} + \kappa_0\rho_c T_c^{-7/2}) & -\frac{21}{2}Q^+\Sigma\kappa_0\rho_c T_c^{-9/2} - 16\sigma T_c^3 & 0 & 3Q^+\Sigma\kappa_0 T_c^{-7/2} \\ 3kT_c - 3\mu m_p H^2 \Omega_K^2 & 3\Sigma k + 8aT_c^3 H \mu m_p & 2aT_c^4 \mu m_p - 6\mu m_p H \Omega_K^2 \Sigma & 0 \\ 3\pi\alpha\Omega_K H^2 & 0 & 6\pi\Sigma\alpha\Omega_K H & 0 \\ -1 & 0 & 2\rho_c & 2H \end{pmatrix}$$

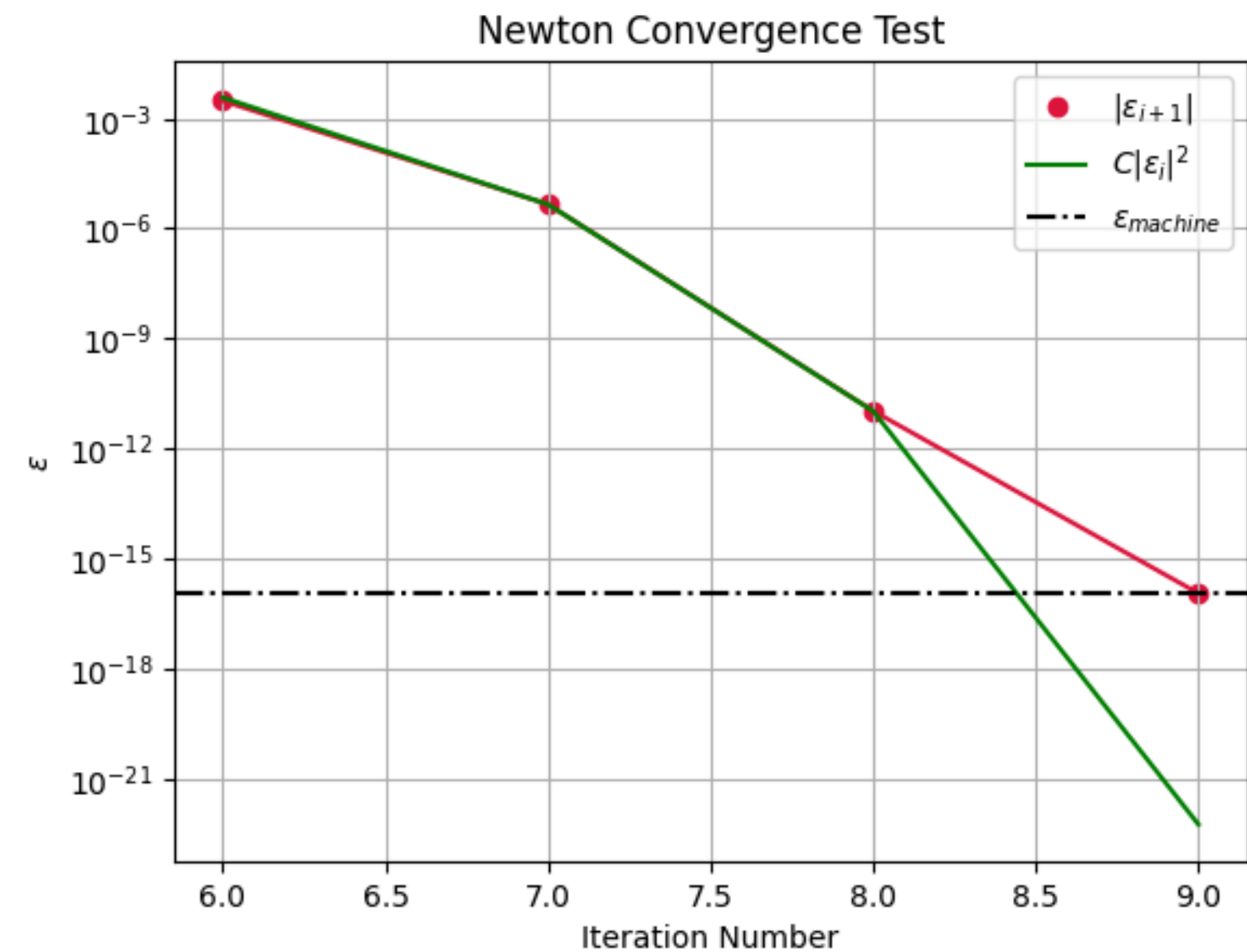
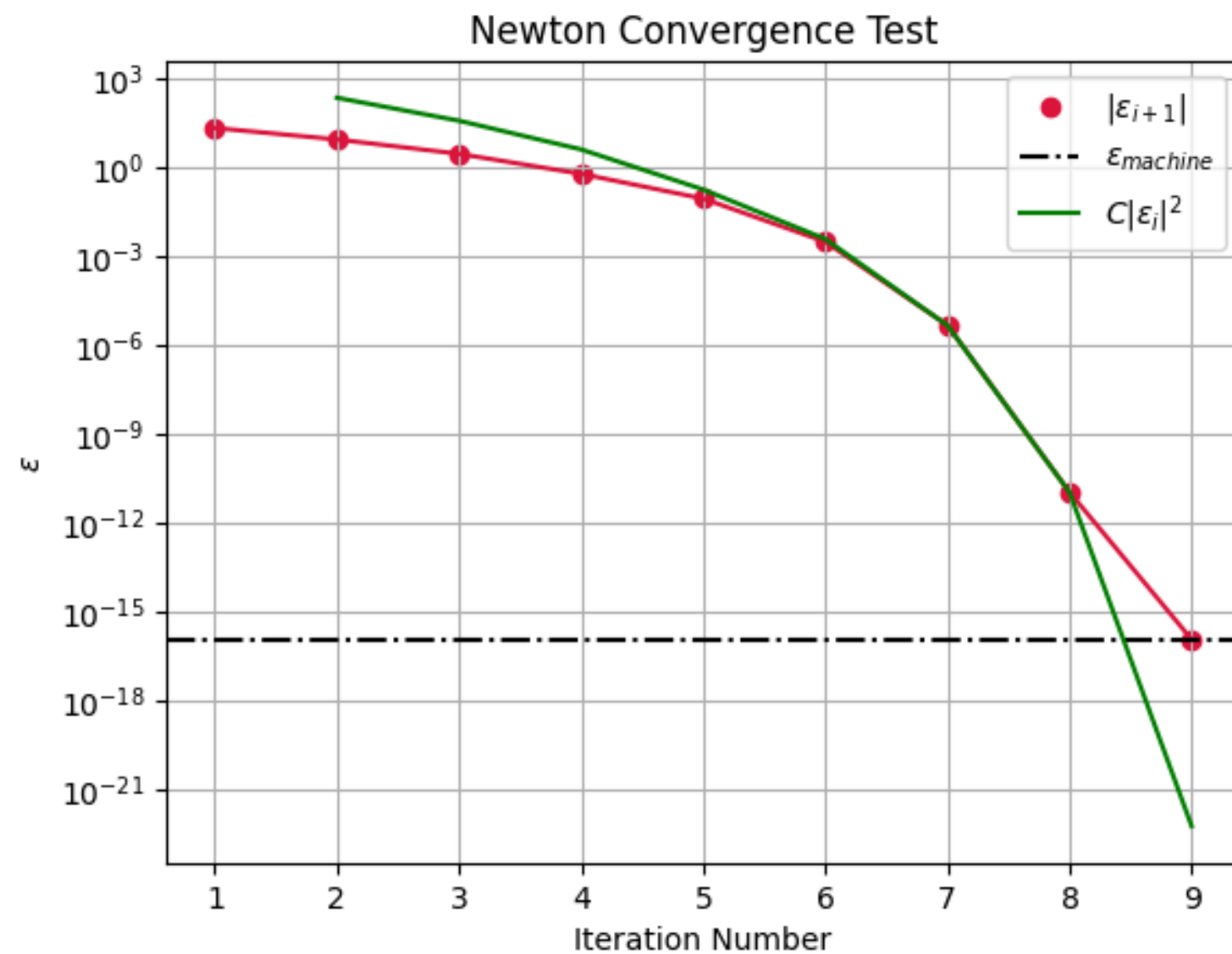
When computing my variables, I had to deal with the enormous difference in magnitude of my physical constants and so I opted to take logarithmic steps.

$$\left\{ \begin{array}{l} c = 3 \times 10^{10} \text{ cm s}^{-1} \\ \kappa_{es} = 0.34 \text{ cm}^2 \text{ g}^{-1} \\ \kappa_0 = 6.4 \times 10^{22} \text{ cm}^5 \text{ g}^{-2} \text{ K}^{7/2} \\ \mu = 0.6 \\ m_p = 1.67 \times 10^{-24} \text{ g} \\ \sigma = 5.67 \times 10^{-5} \text{ erg cm}^{-2} \text{ s}^{-1} \text{ K}^{-4} \end{array} \right. \quad \left\{ \begin{array}{l} a = 7.56 \times 10^{-15} \text{ erg cm}^{-3} \text{ K}^{-4} \\ k = 1.38 \times 10^{-16} \text{ erg K}^{-1} \\ M_\odot = 2 \times 10^{33} \text{ g} \\ \eta = 1 - \frac{\sqrt{8}}{3} \approx 0.057 \\ \sigma_T = 6.652 \times 10^{-25} \text{ cm}^2 \\ h = 6.626 \times 10^{-27} \text{ erg s} \end{array} \right.$$

```
def new_step_log(yk,R,Rstar,M,Mdot,alpha):
    xk=np.exp(yk)
    F_val=F(xk,R,Rstar,M,Mdot,alpha)
    J=np.array(Jacobian(xk,R,Rstar,M,Mdot,alpha))
    logJ=J*xk
    delta_yk=Gauss_Jordan(logJ.tolist(),F_val)
    return delta_yk
```

Newton-Rhapon

Newton-Rhapon has quadratic convergence $\rightarrow |\epsilon_{j+i}| = C|\epsilon_j|^2$



Gauss-Jordan

Gauss-Jordan is linear systems solver:

1. Combines coefficients and constants;
2. Looks for top left pivot;
3. Makes it 1;
4. Clears the column;
5. Iterates on the following column;
6. Solution builds up in the last column.

- Number of iterations goes as N^3
- Roundoff error

```
def Gauss_Jordan(A,b):
    n=len(A)
    C=[A[i]+[b[i]] for i in range(n)]
    eps=1e-15

    for i in range(n):
        max_row=i
        for r in range(i+1,n):
            if np.abs(C[r][i]) > np.abs(C[max_row][i]):
                max_row = r

        if max_row!=i:
            C[i],C[max_row]=C[max_row],C[i]

        if np.abs(C[i][i])<eps:
            return [0]*n

        diag=C[i][i]
        for c in range(i,n+1):
            C[i][c]/=diag

        for r in range(n):
            if r!=i:
                mult=C[r][i]
                for c in range(i,n+1):
                    C[r][c]-=mult*C[i][c]

    return [row[-1] for row in C]
```

Simpson

Simpson is an integration method of the Newton-Cotes family

$$\int_a^b f(x)dx \approx \sum_{i=1}^N Q_i \quad Q_i = \frac{h_i}{6} \left(f(a_i) + 4f(m_i) + f(b_i) \right) + O(h_i^5)$$

I used a logarithmic step-size to adapt to the magnitudes of my problem, since I'm spacing over a high range of radii.

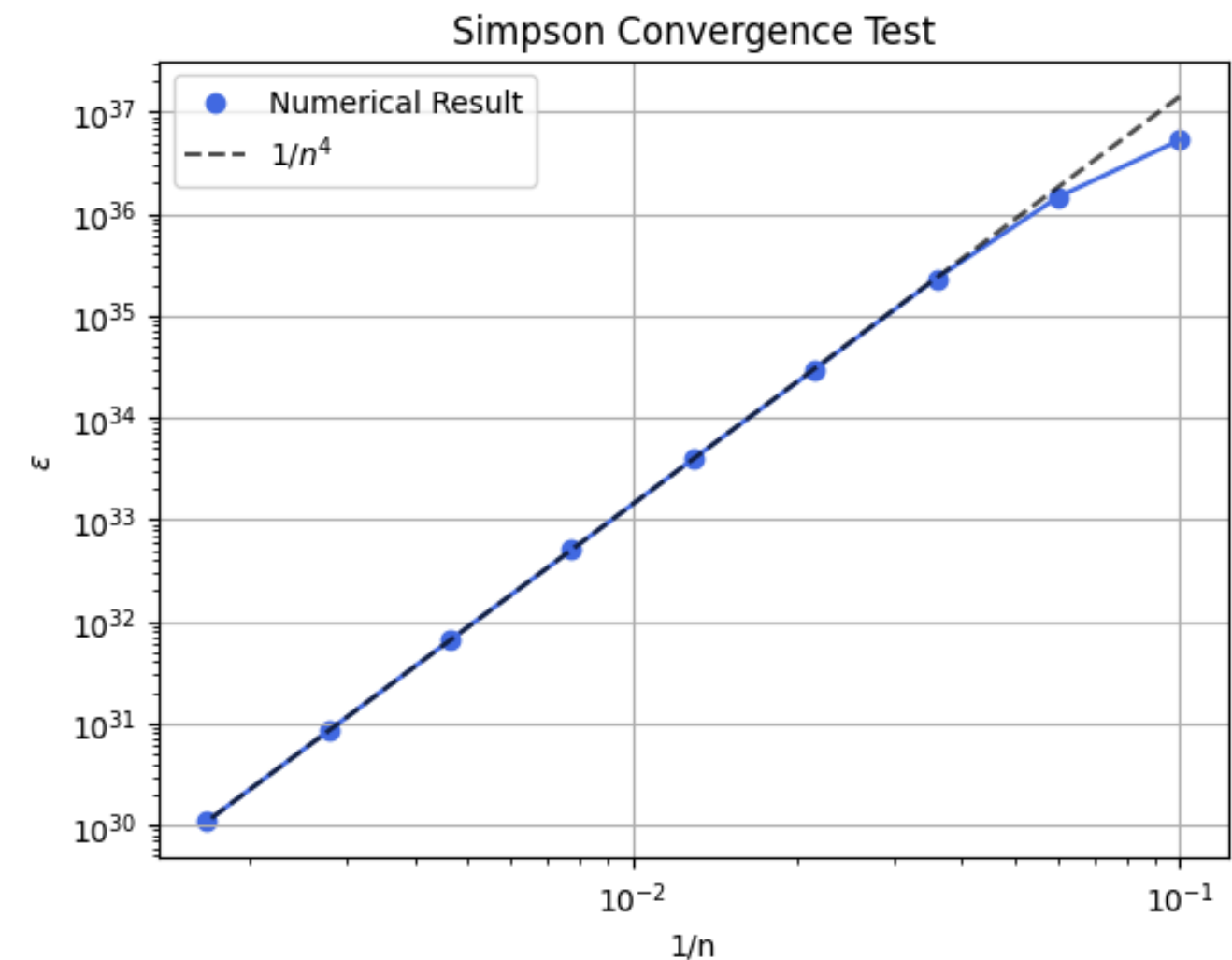
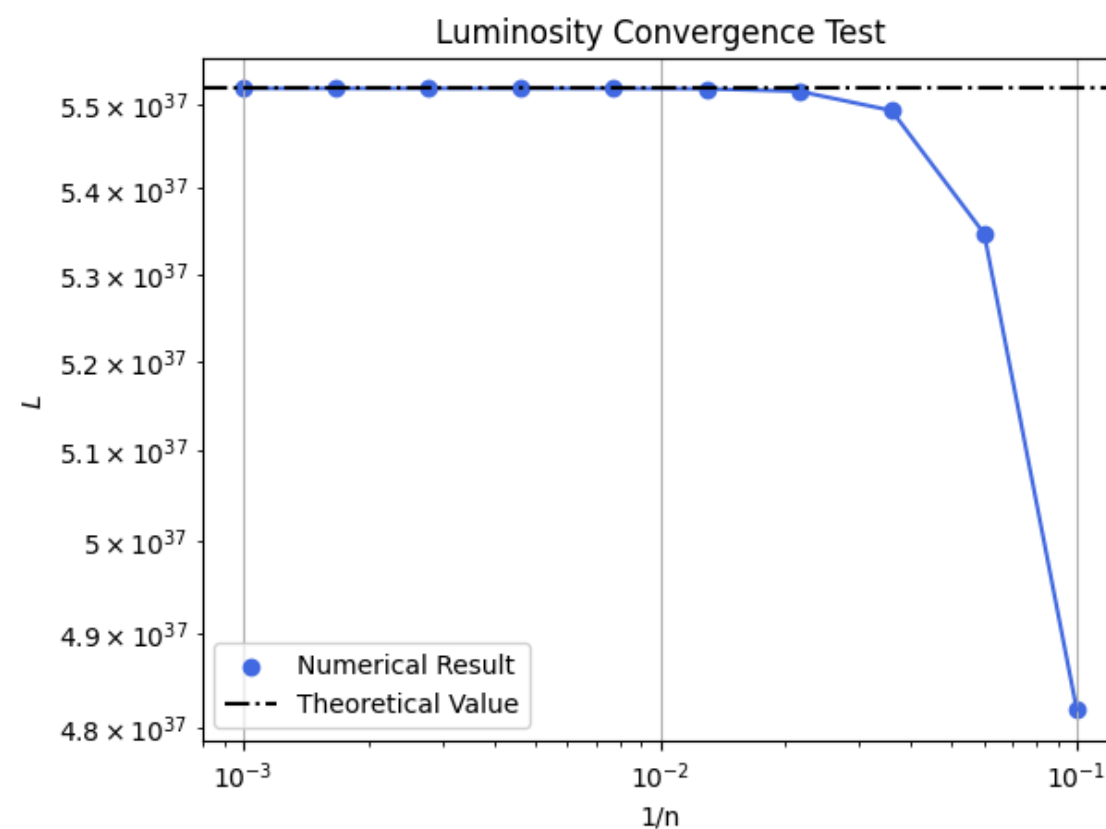
```
def Simpson(a,b,n):
    logR=np.logspace(np.log10(a),np.log10(b),n+1)
    h=np.diff(logR)
    left=logR[:-1]
    right=logR[1:]
    m=(left+right)/2
    Q=0
    for i in range(n):
        Q+=h[i]/6*(dL_dR(left[i],Rstar,M,Mdot)+dL_dR(right[i],Rstar,M,Mdot)+4*dL_dR(m[i],Rstar,M,Mdot))
    return Q
```

Simpson

$$Q_i = \frac{h_i}{6} \left(f(a_i) + 4f(m_i) + f(b_i) \right) + O(h_i^5)$$

Since the grid is logarithmic I couldn't use the classic check: $O(h^4)$

But that relation still holds for the number of points in the grid, so I would expect: $O(1/n^4)$



Radial Profiles

Region A (Inner)

- Radiation Pressure + Electron Scattering
- $\Sigma \propto R^{3/2}$
- $T_c \propto R^{-3/8}$
- $T_{\max} \propto \dot{M}^0$

Region B (Mid)

- Gas Pressure + Electron Scattering
- $\Sigma \propto R^{-3/5}$
- $T_c \propto R^{-9/10}$
- $T_{\max} \propto \dot{M}^{2/5}$

Region C (Outer)

- Gas Pressure + Kramers' Opacity
- $\Sigma \propto R^{-3/4}$
- $T_c \propto R^{-3/4}$
- $T_{\max} \propto \dot{M}^{3/10}$

$$f(R) \equiv 1 - \left(\frac{R_*}{R} \right)^{1/2}$$

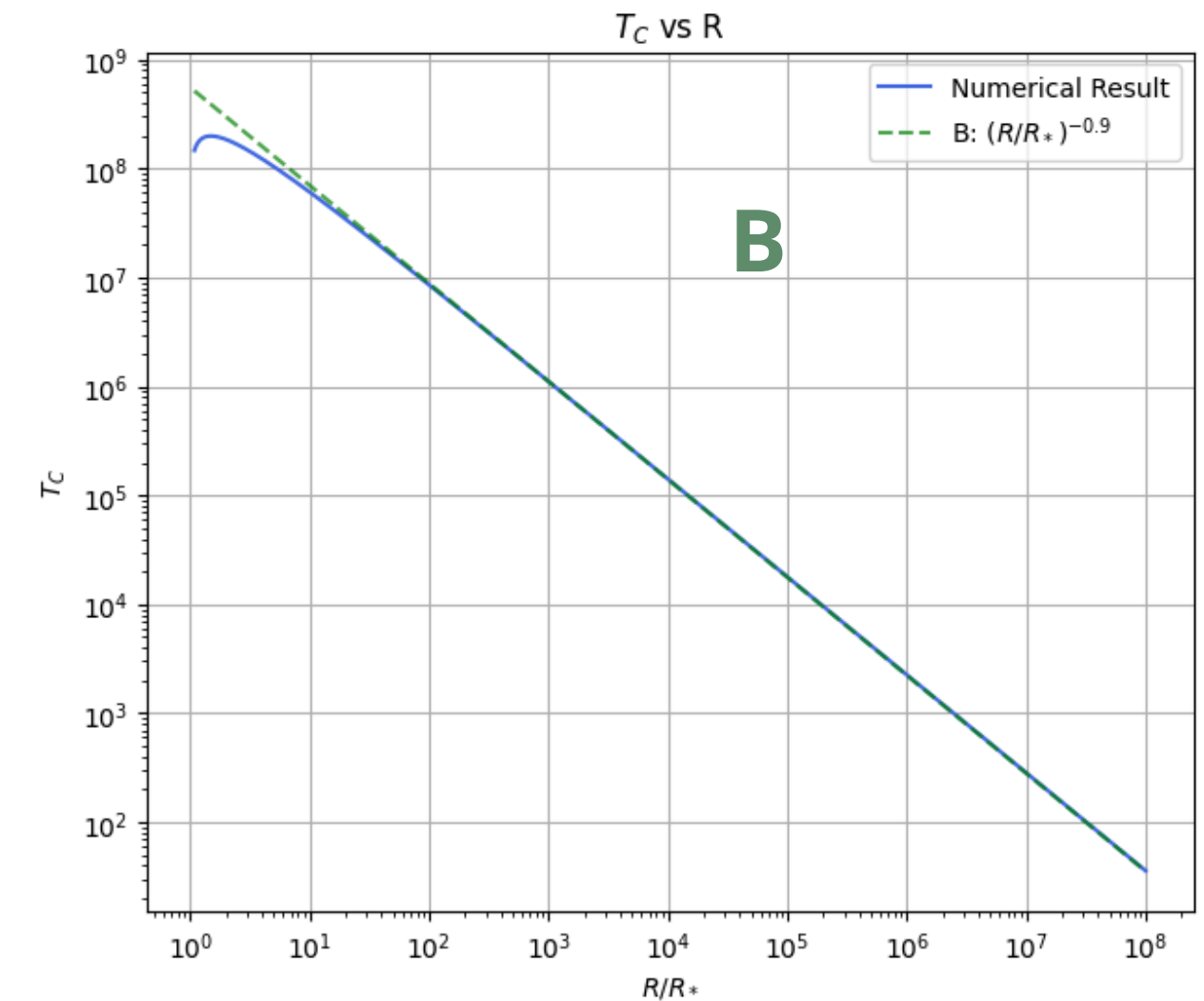
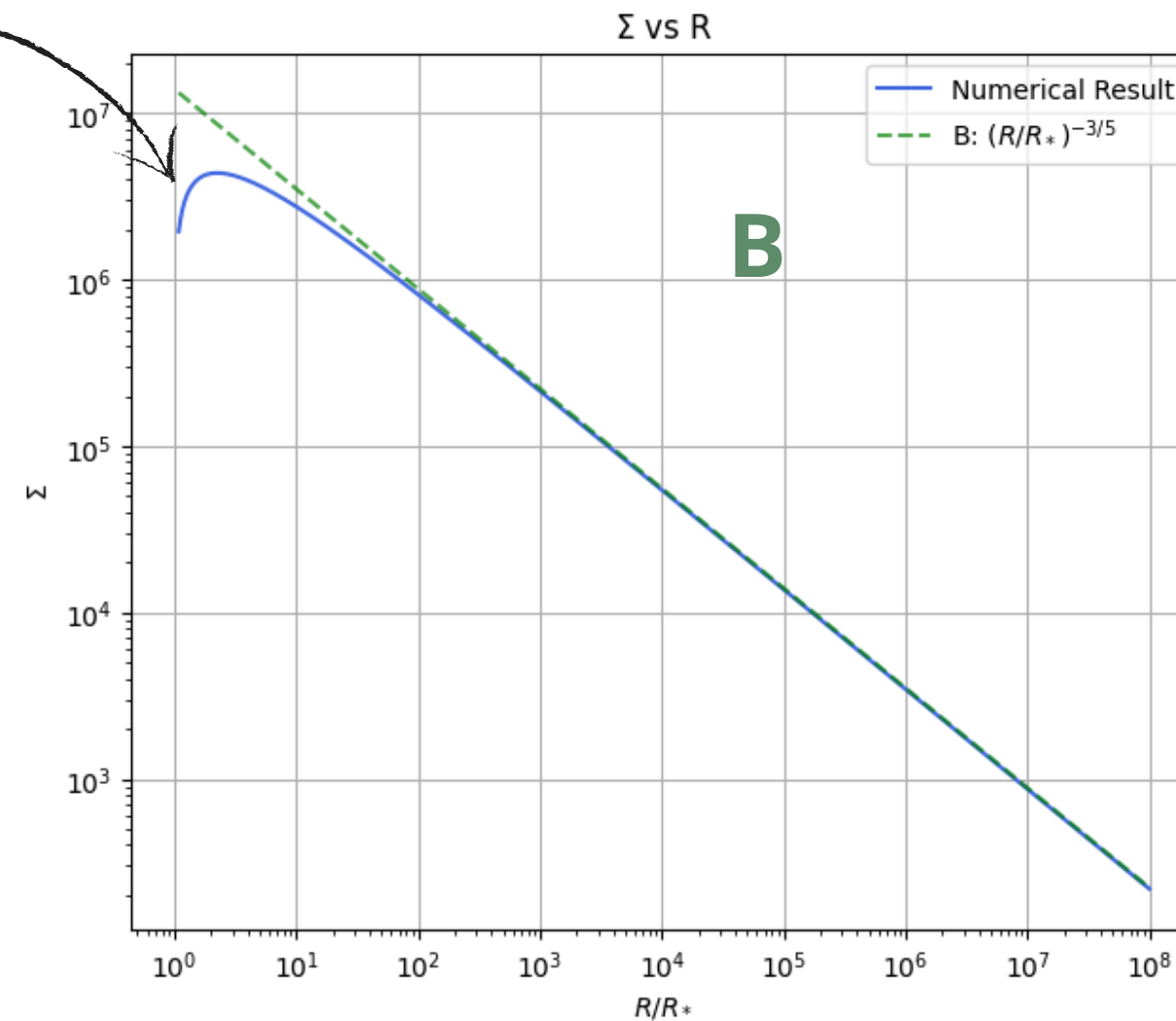
The $f(R)$ zero-torque term becomes relevant when R is close to the ISCO, producing unphysical effects at the beginning of Region A

Due to this approximation, my analysis starts at slightly larger radii to avoid encountering the divergence

Initial Guess

Zero-Torque
Approximation

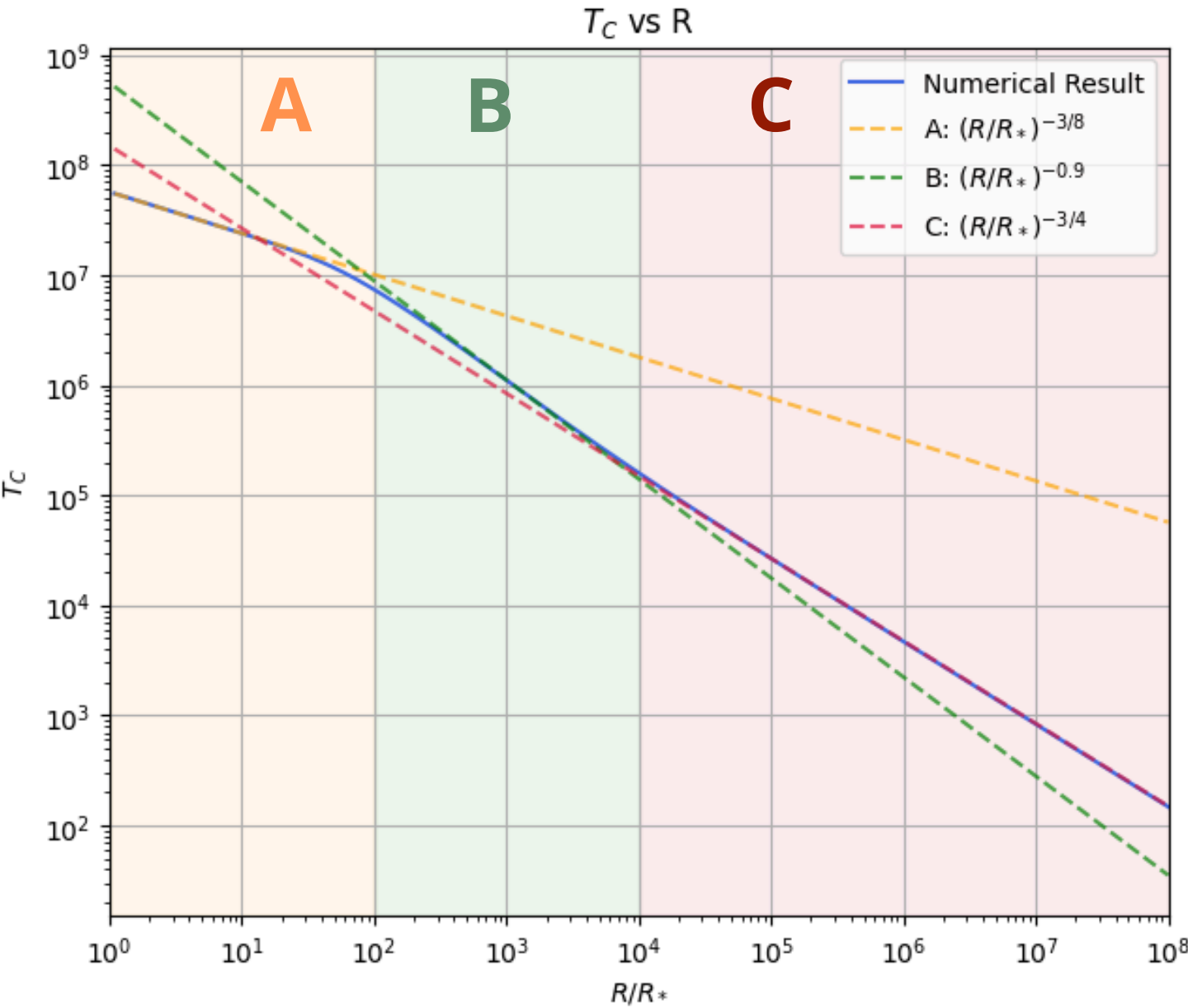
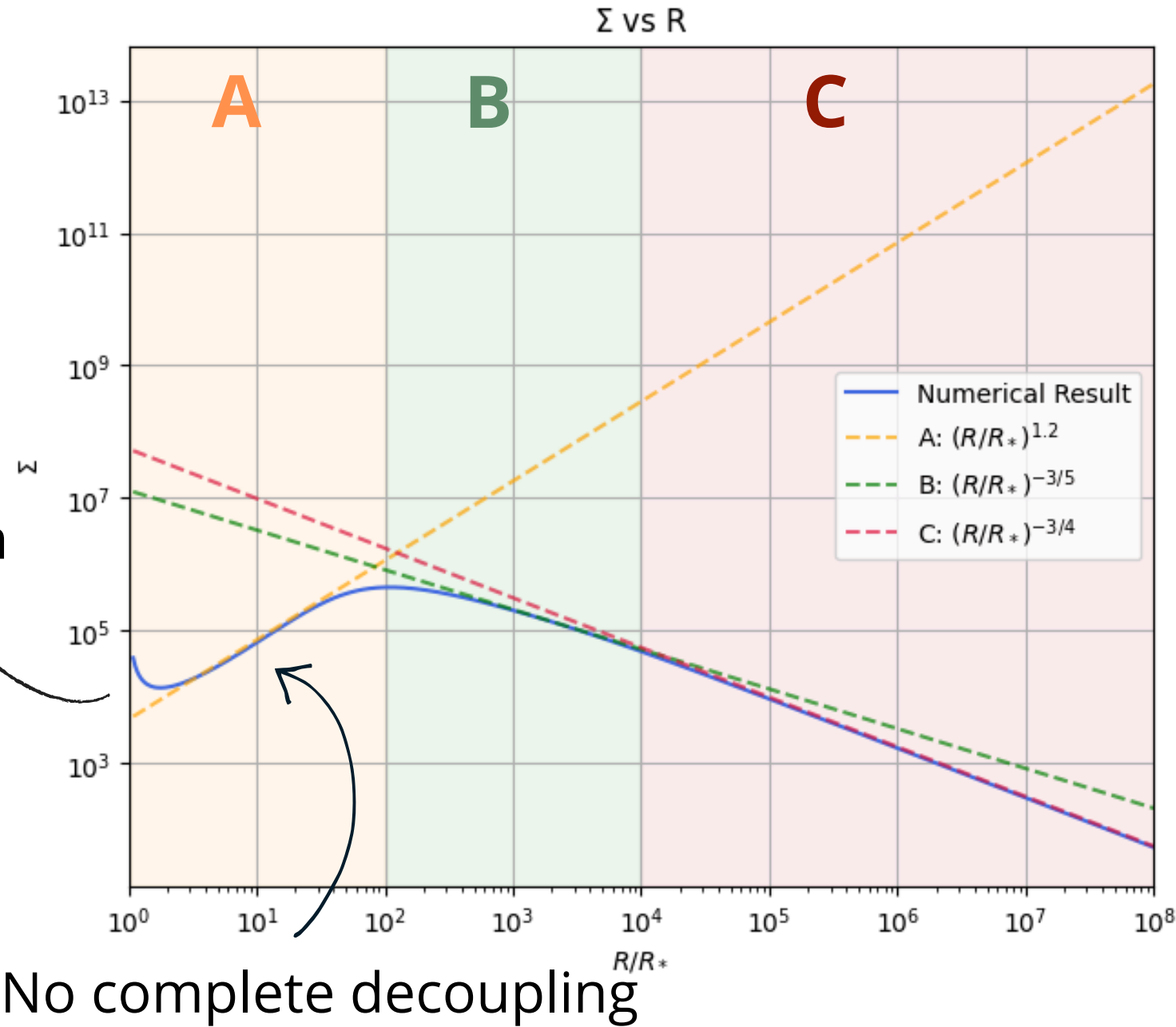
$$\begin{cases} \alpha = 0.001 \\ M = 10M_{\odot} \\ R_* = \frac{6GM}{c^2} \\ \dot{M} = 0.1\dot{M}_{Edd} \end{cases}$$



Σ and T_c

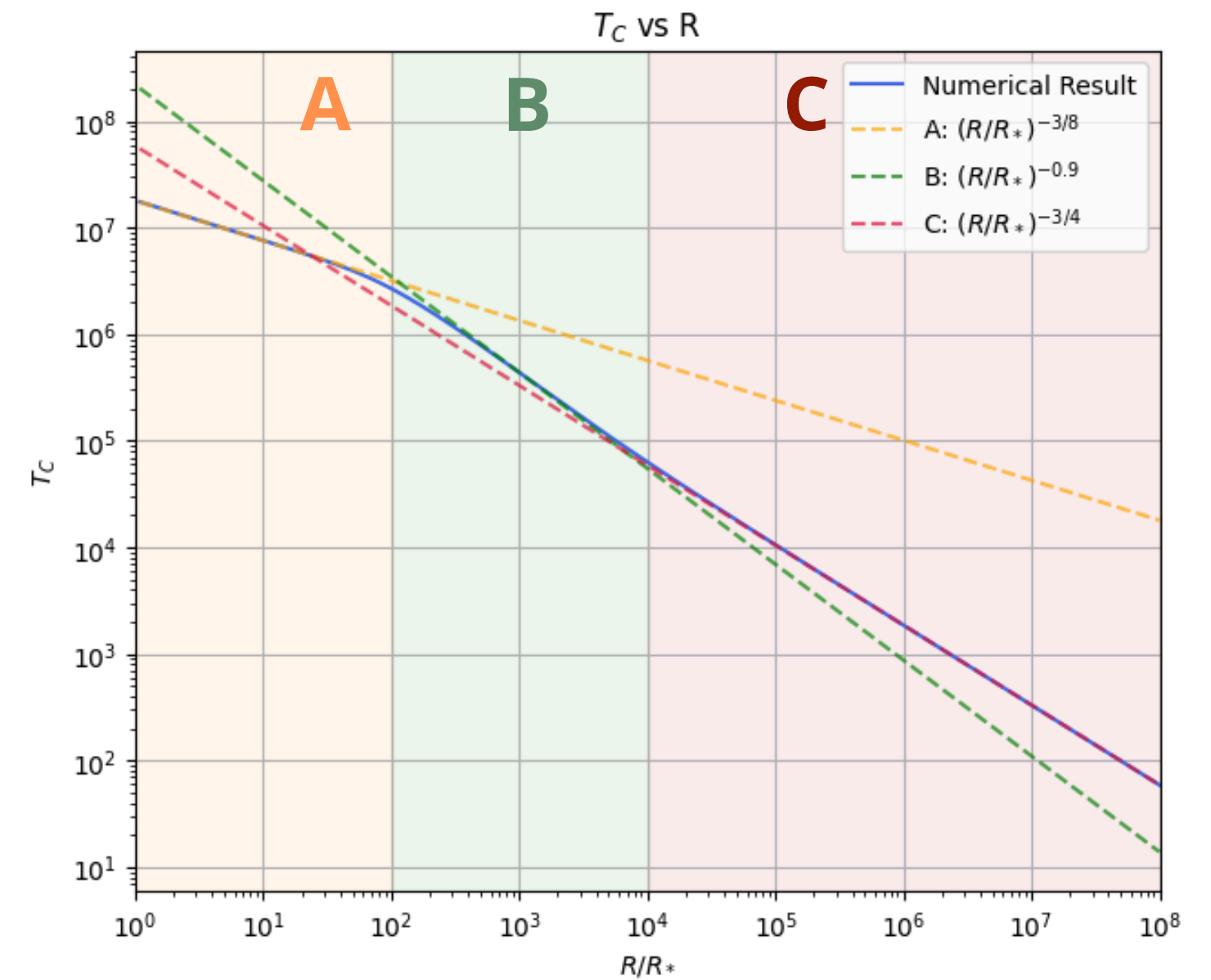
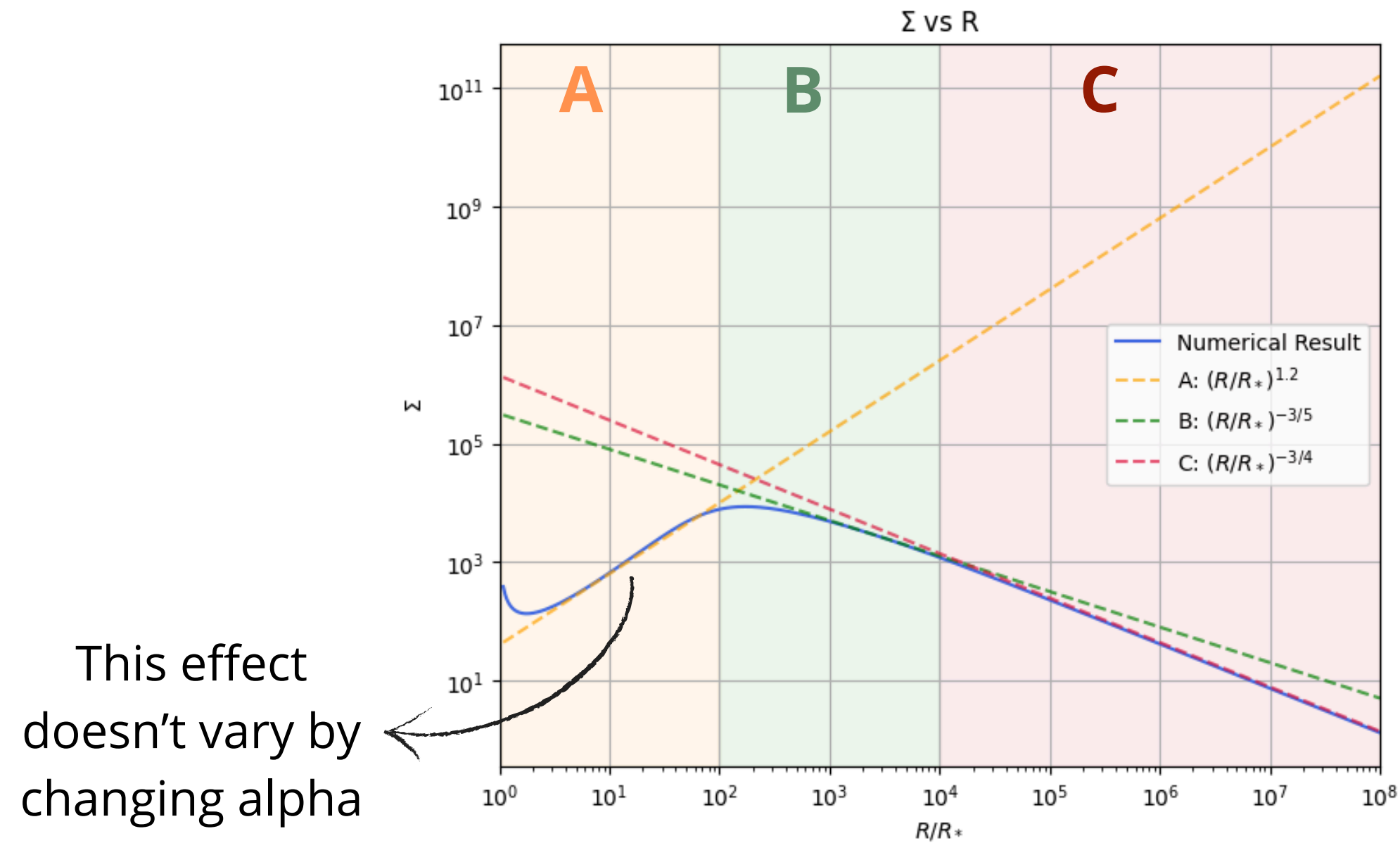
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Zero-Torque
Approximation



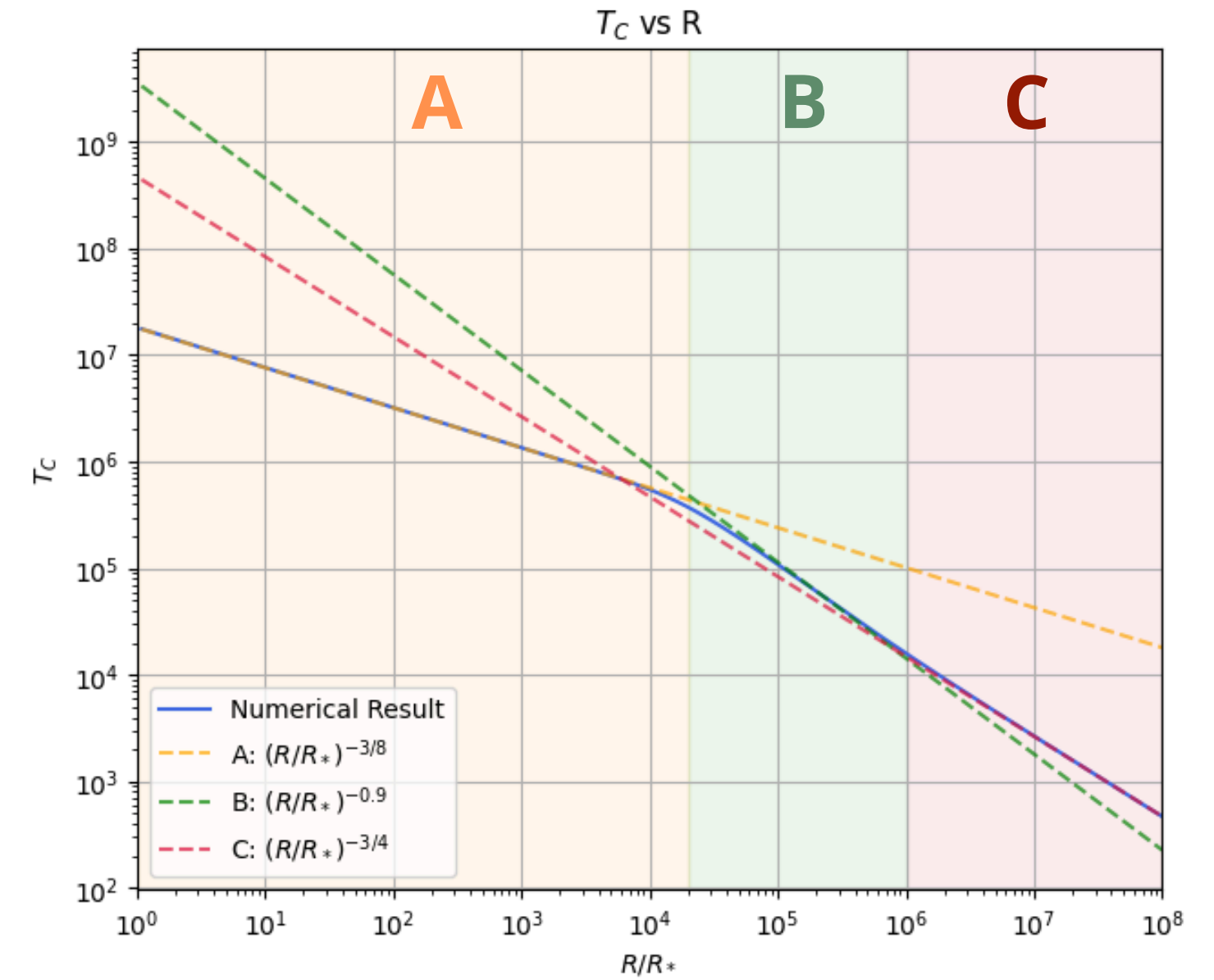
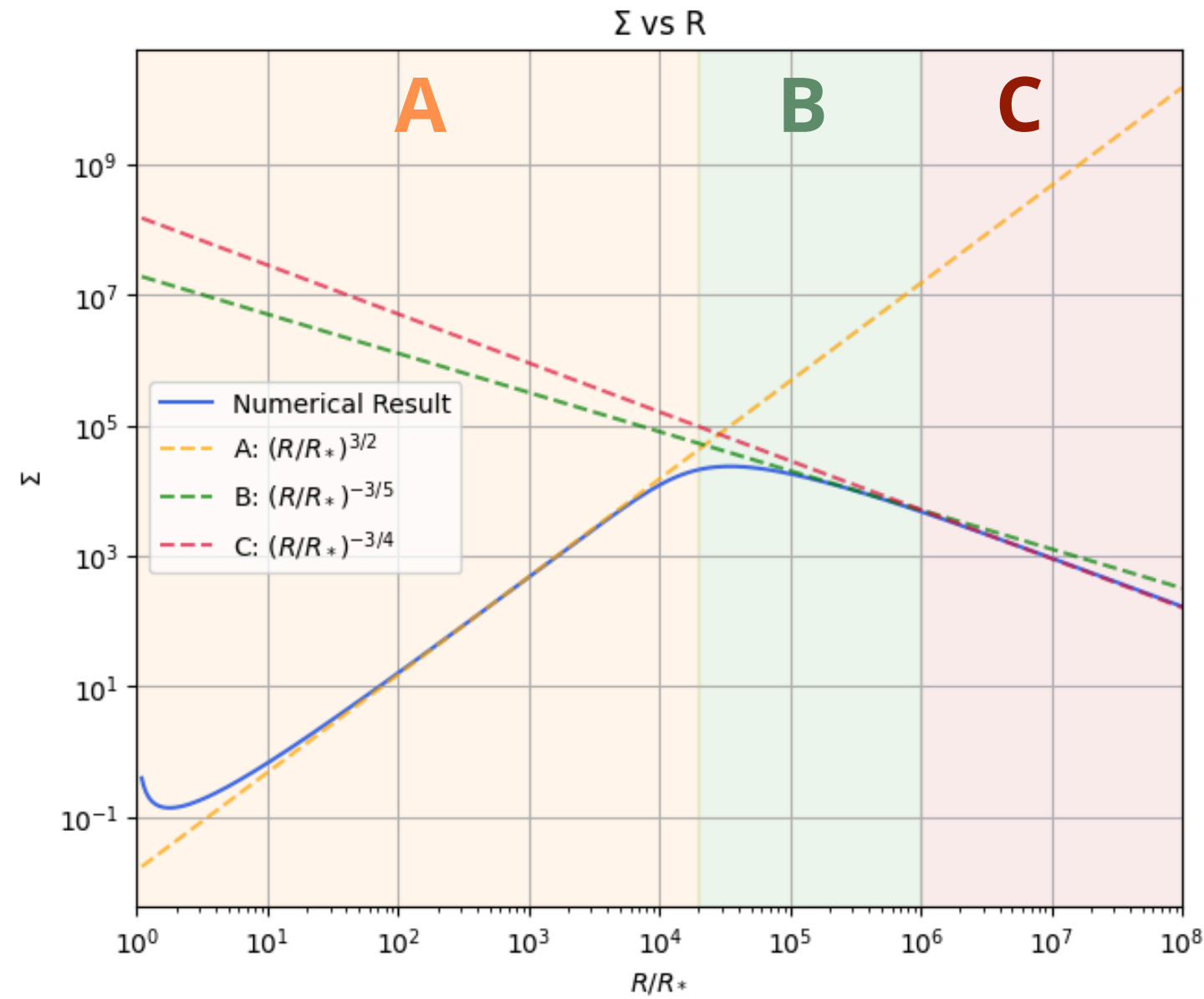
Σ and T_c

$$\begin{cases} \alpha = 0.1 \\ M = 10M_{\odot} \\ \dot{M} = 0.1\dot{M}_{Edd} \end{cases}$$



Σ and T_c

$$\begin{cases} \alpha = 0.1 \\ M = 10M_{\odot} \\ \dot{M} = 10^2 \dot{M}_{Edd} \end{cases}$$

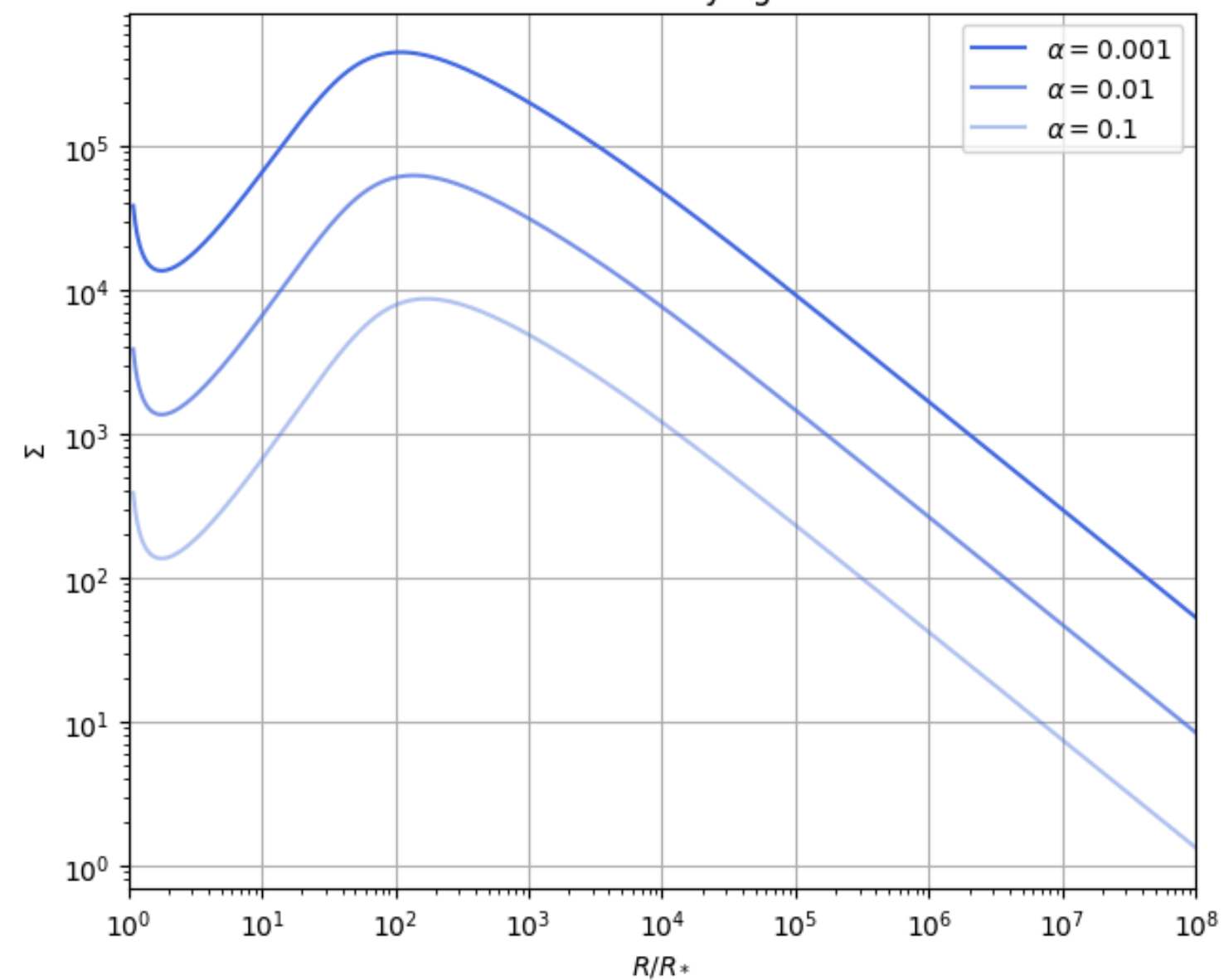


$$M = 10M_{\odot}$$

Σ

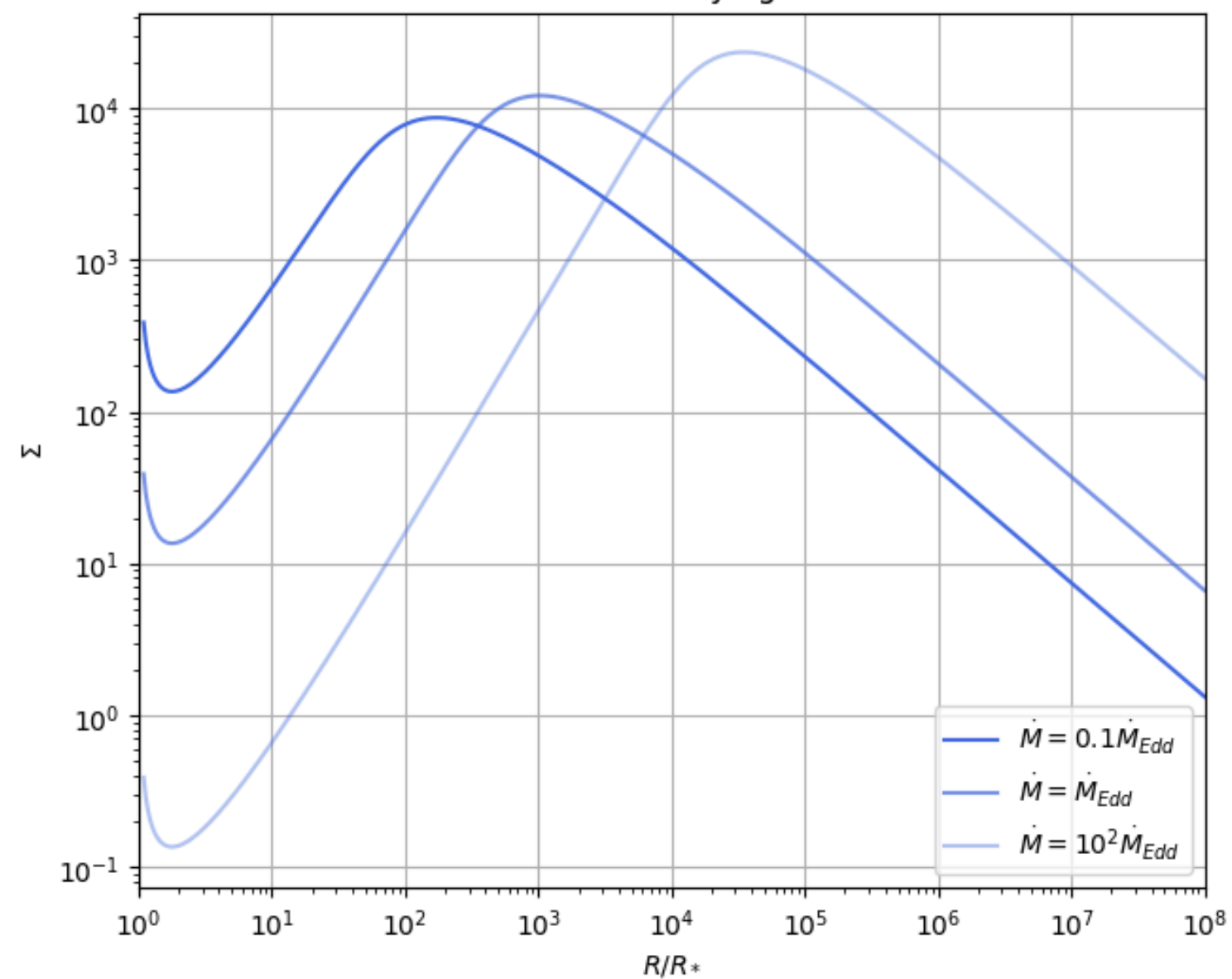
$$\dot{M} = 0.1\dot{M}_{Edd}$$

Σ vs R - Varying α



$$\alpha = 0.1$$

Σ vs R - Varying $\dot{M}$

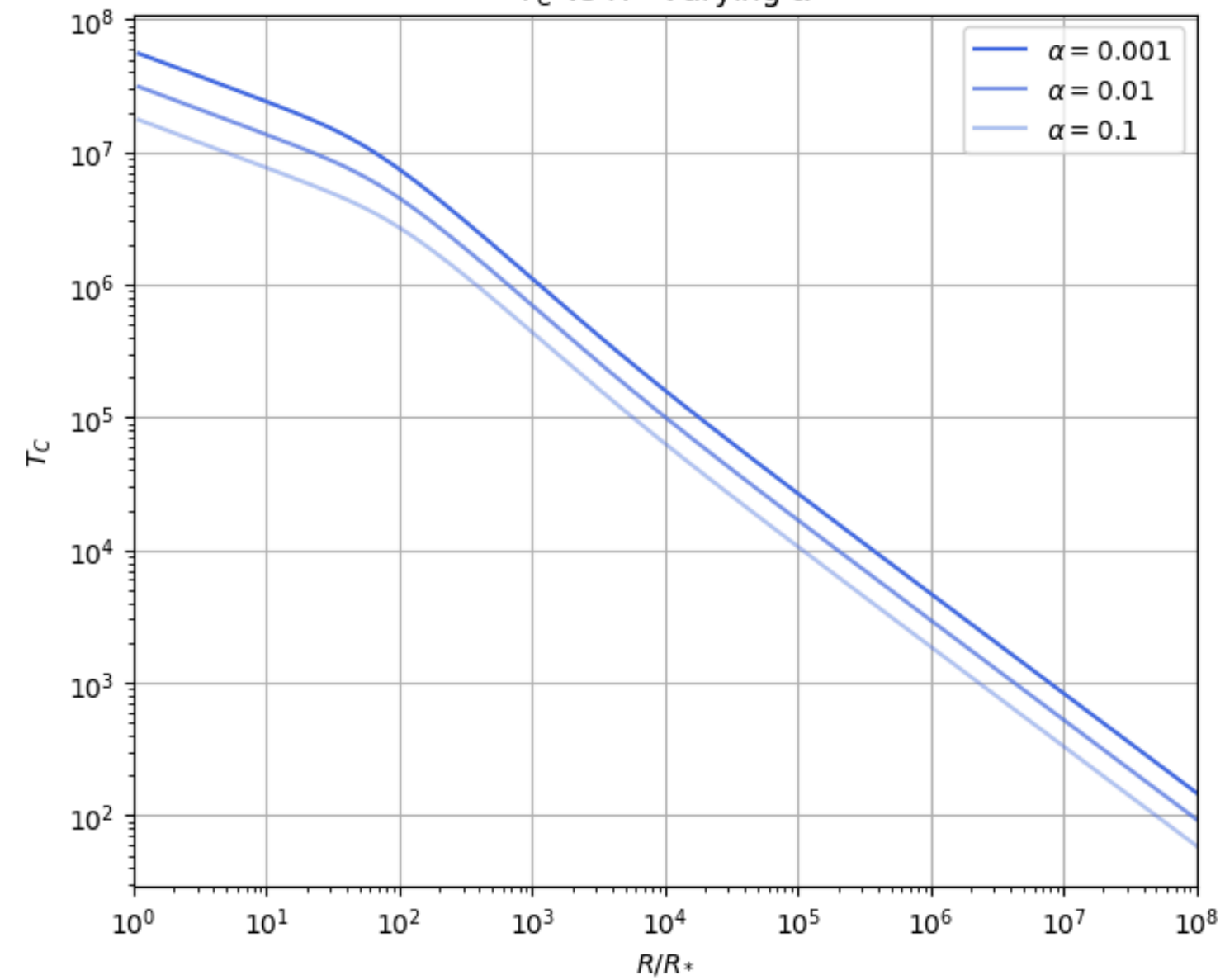


$$M = 10M_{\odot}$$

T_c

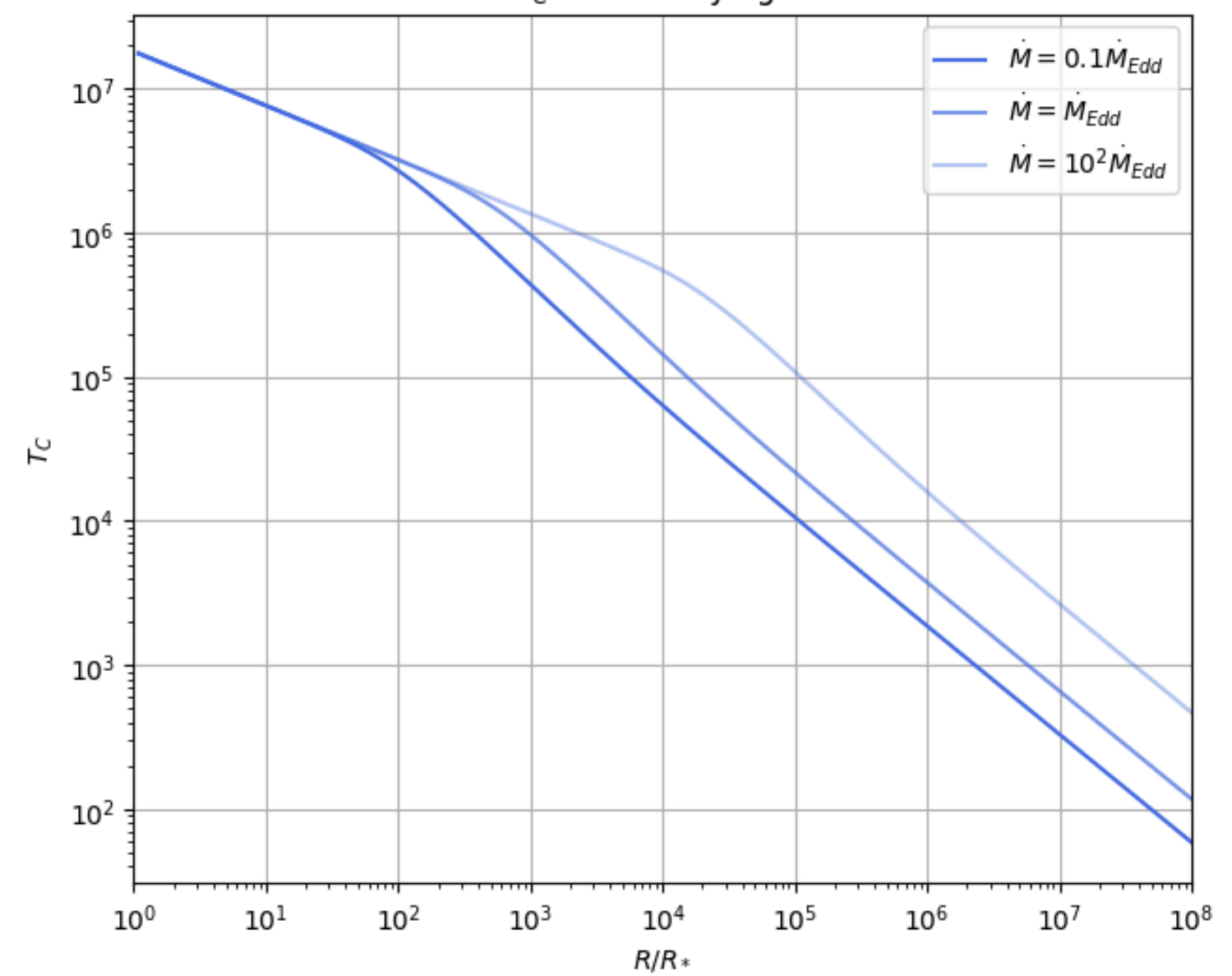
$$\dot{M} = 0.1\dot{M}_{Edd}$$

T_c vs R - Varying α



$$\alpha = 0.1$$

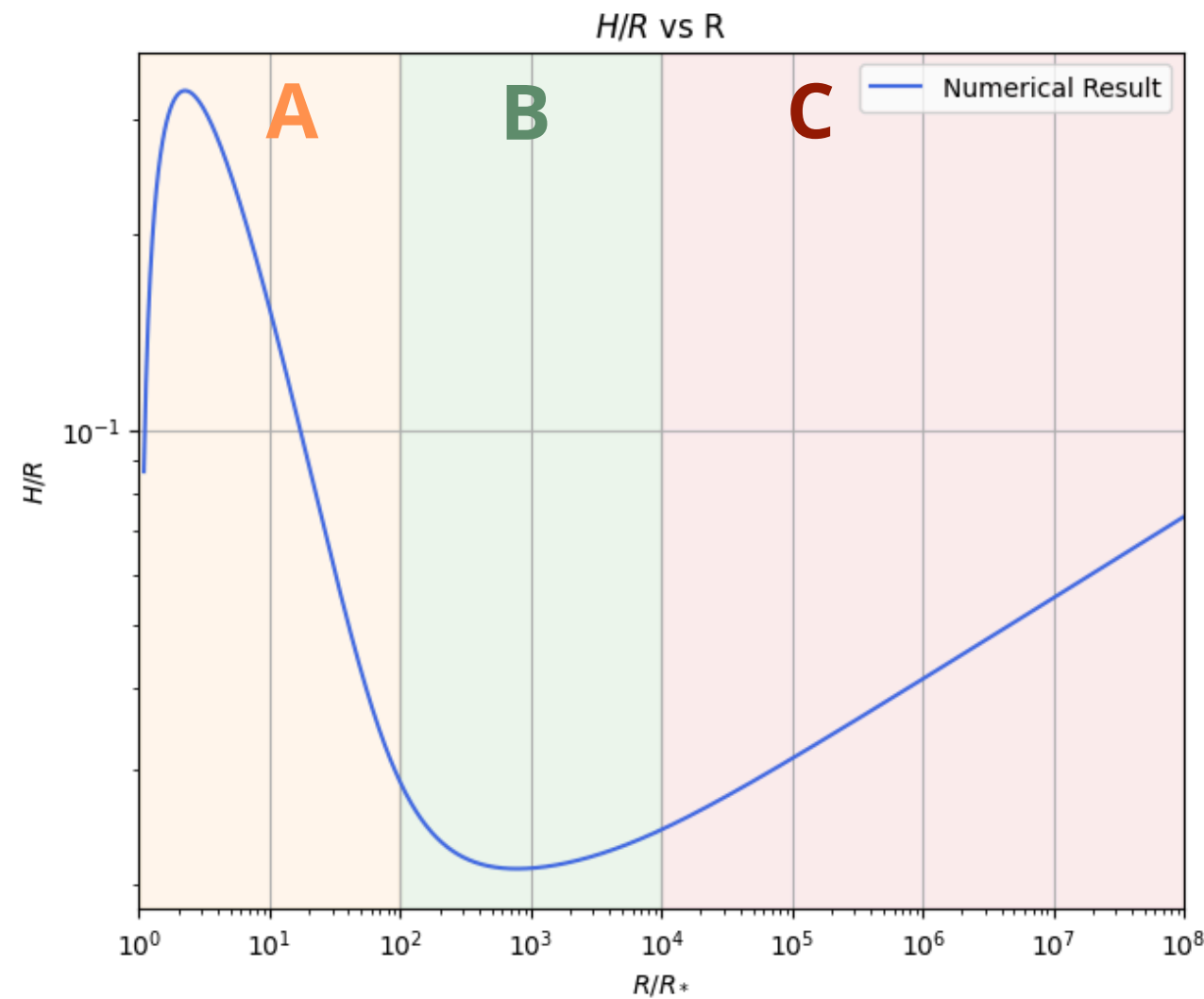
T_c vs R - Varying $\dot{M}$



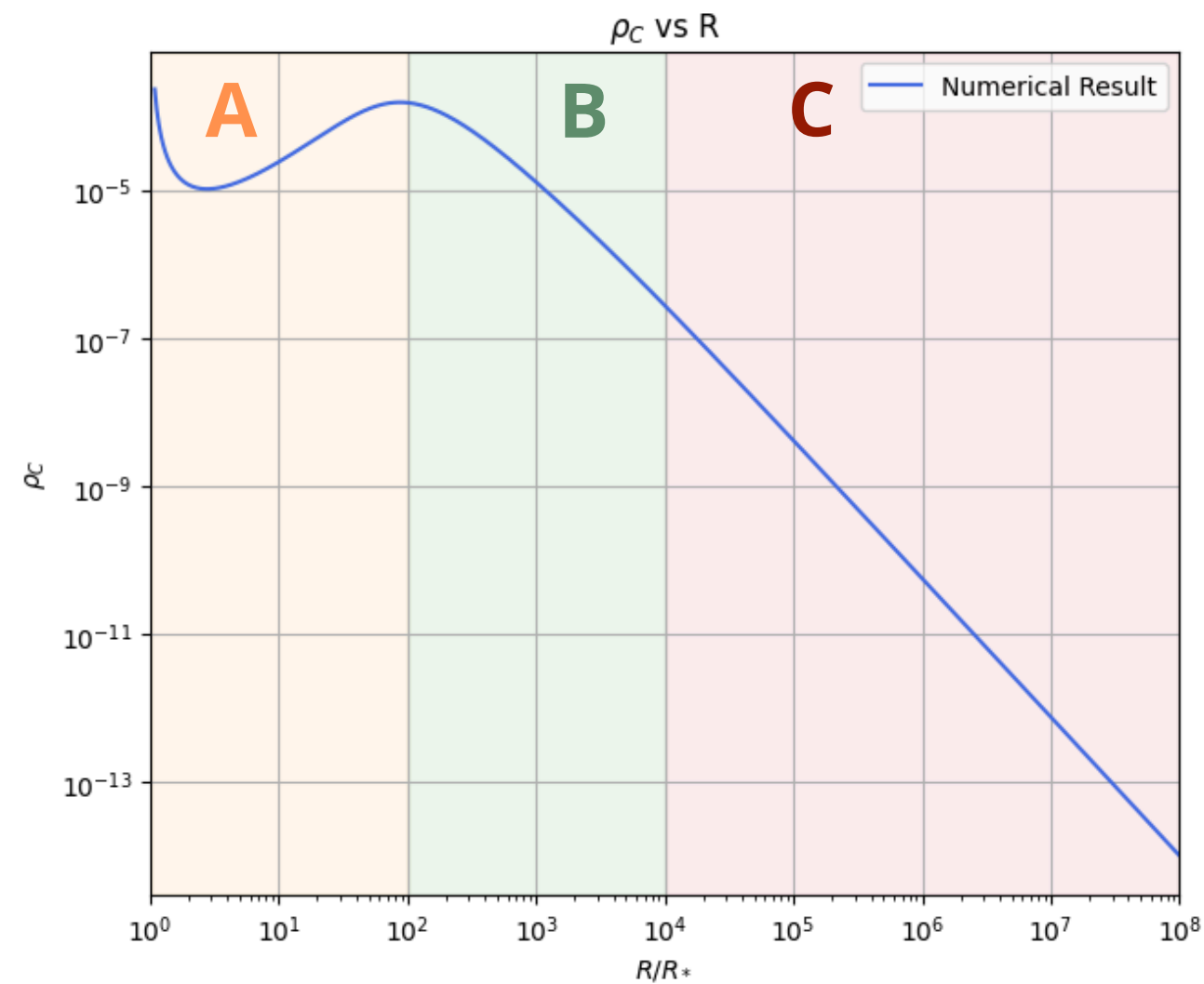
H/R and ρ_c

$$\begin{cases} \alpha = 0.1 \\ M = 10M_{\odot} \\ \dot{M} = 0.1\dot{M}_{Edd} \end{cases}$$

We can notice that the disk is thin in the radiation pressure dominated regime and then enlarges going outwards

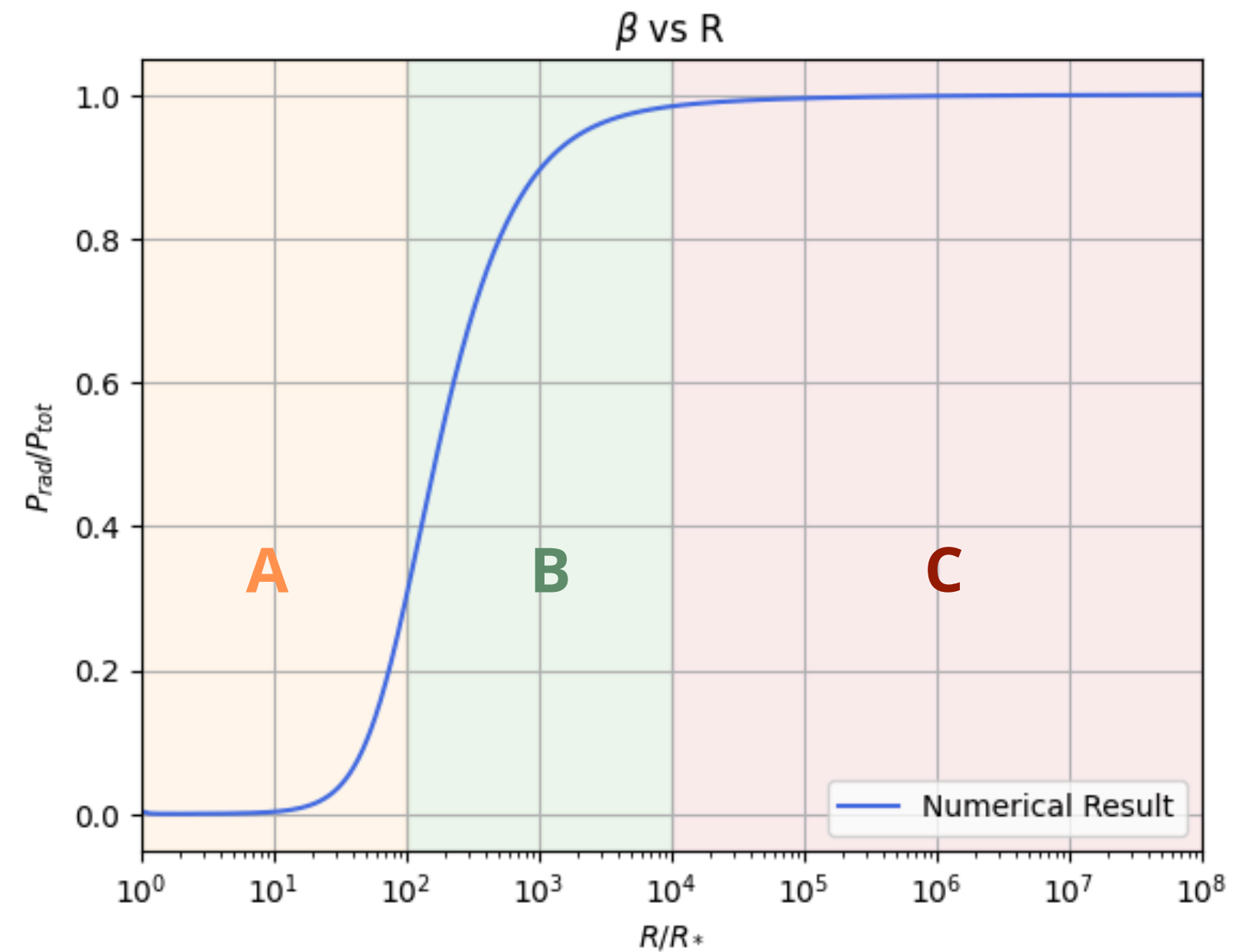
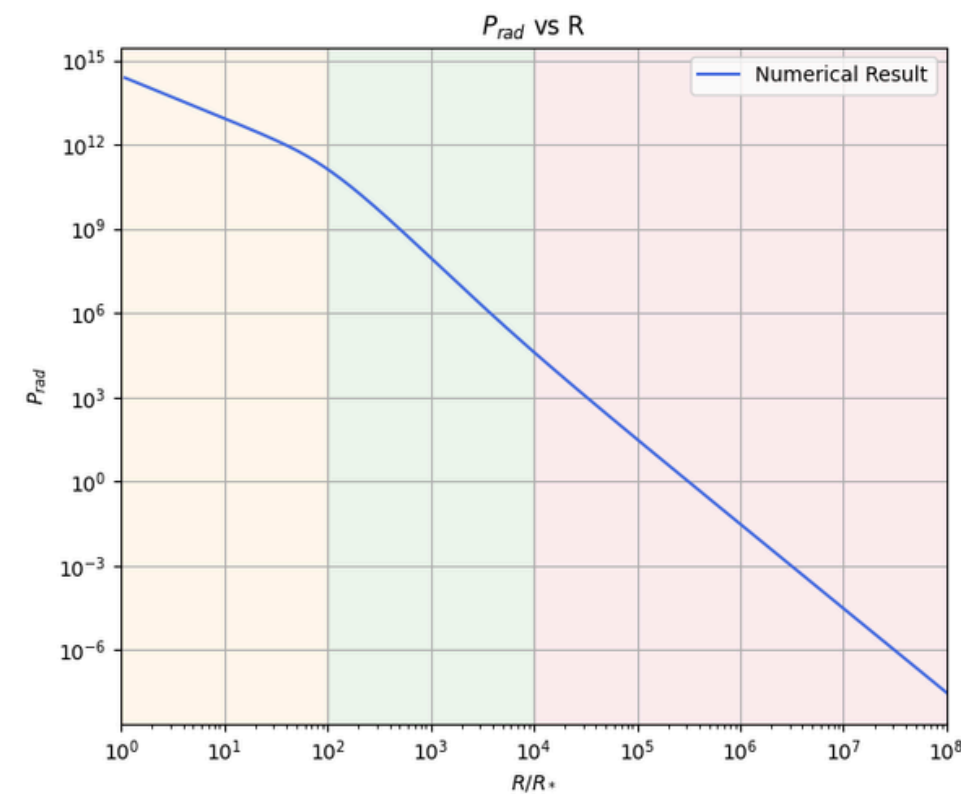
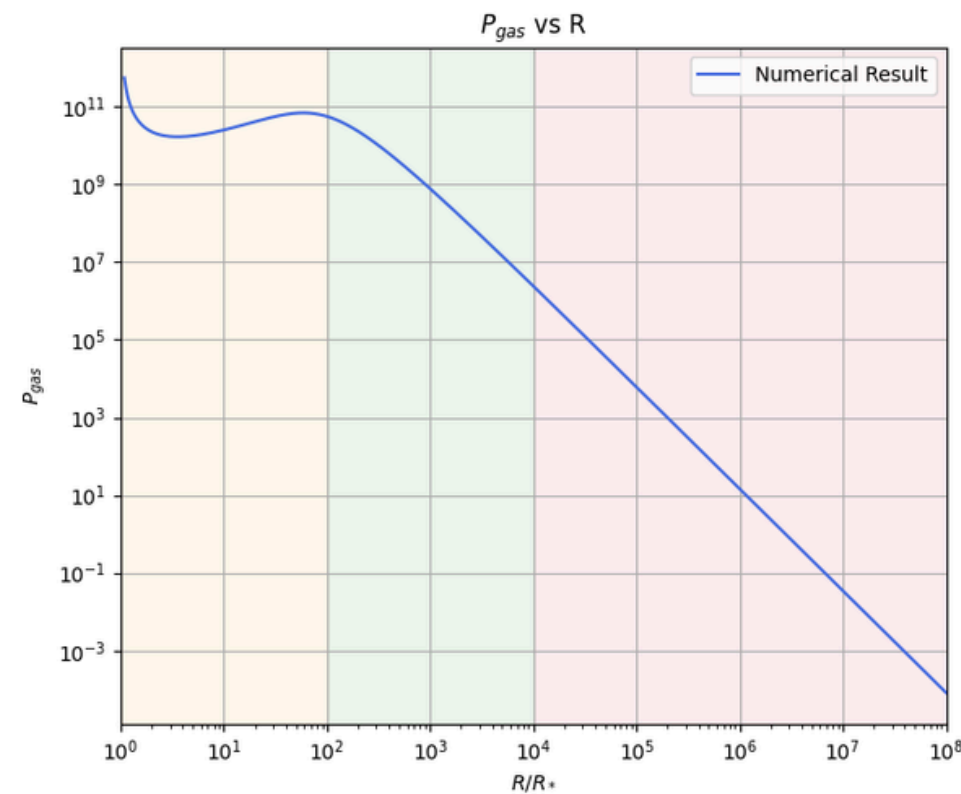


We can notice the density having a similar behaviour to the surface density, it grows in region A and then drops once gas pressure becomes dominant



P_{gas} , P_{rad} and β

$$\begin{cases} \alpha = 0.1 \\ M = 10M_{\odot} \\ \dot{M} = 0.1\dot{M}_{Edd} \end{cases}$$



Radiation pressure starts higher, but drops faster. We can notice the slope change following a similar behaviour to the temperature

$$P_c = \frac{\rho_c k T_c}{\mu m_p} + \frac{a T_c^4}{3}$$

Peak Temperature

$$\begin{cases} \alpha = 0.1 \\ M = 10M_{\odot} \end{cases}$$

Region A (Inner)

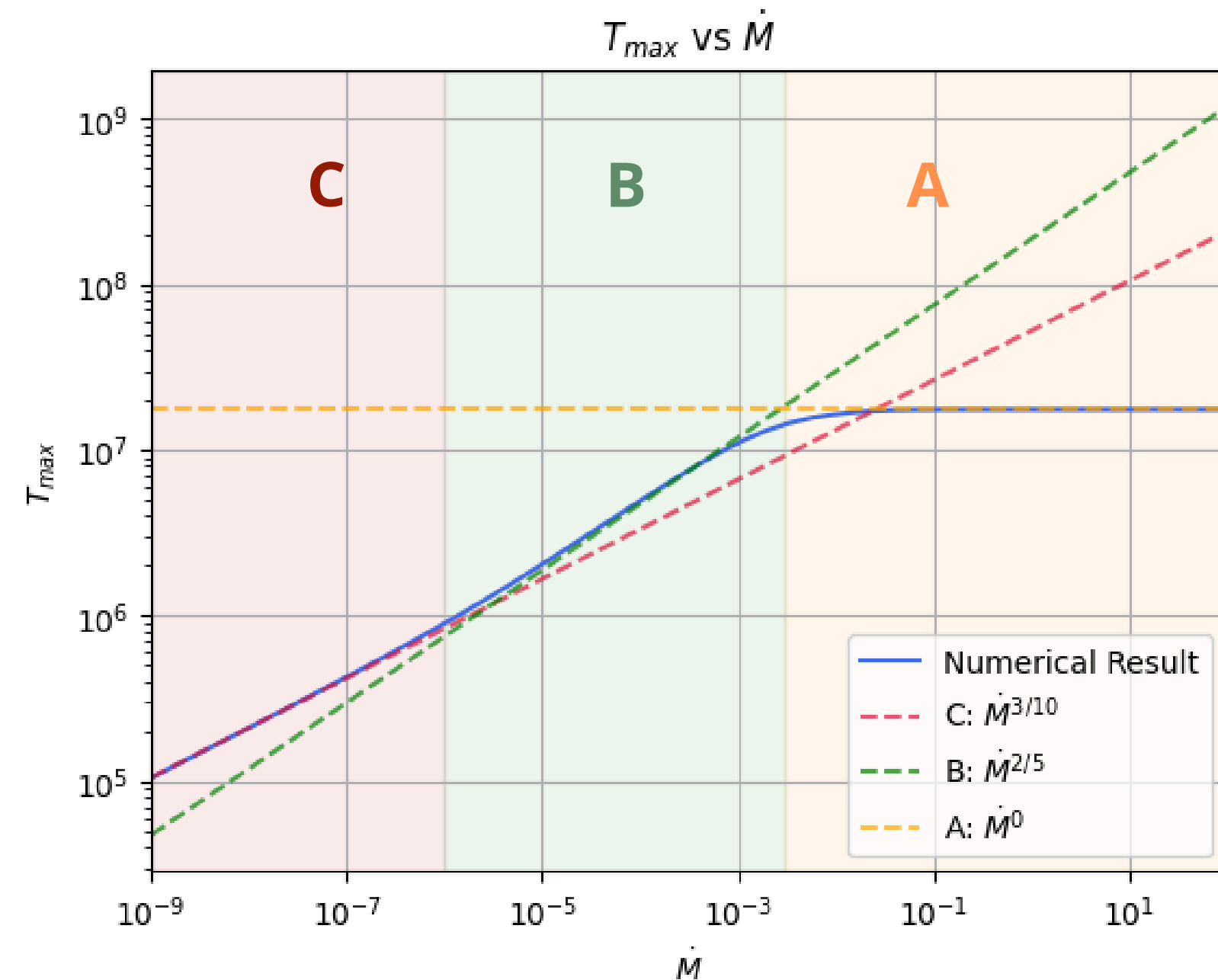
- Radiation Pressure + Electron Scattering
- $\Sigma \propto R^{3/2}$
- $T_c \propto R^{-3/8}$
- $T_{\max} \propto \dot{M}^0$

Region B (Mid)

- Gas Pressure + Electron Scattering
- $\Sigma \propto R^{-3/5}$
- $T_c \propto R^{-9/10}$
- $T_{\max} \propto \dot{M}^{2/5}$

Region C (Outer)

- Gas Pressure + Kramers' Opacity
- $\Sigma \propto R^{-3/4}$
- $T_c \propto R^{-3/4}$
- $T_{\max} \propto \dot{M}^{3/10}$



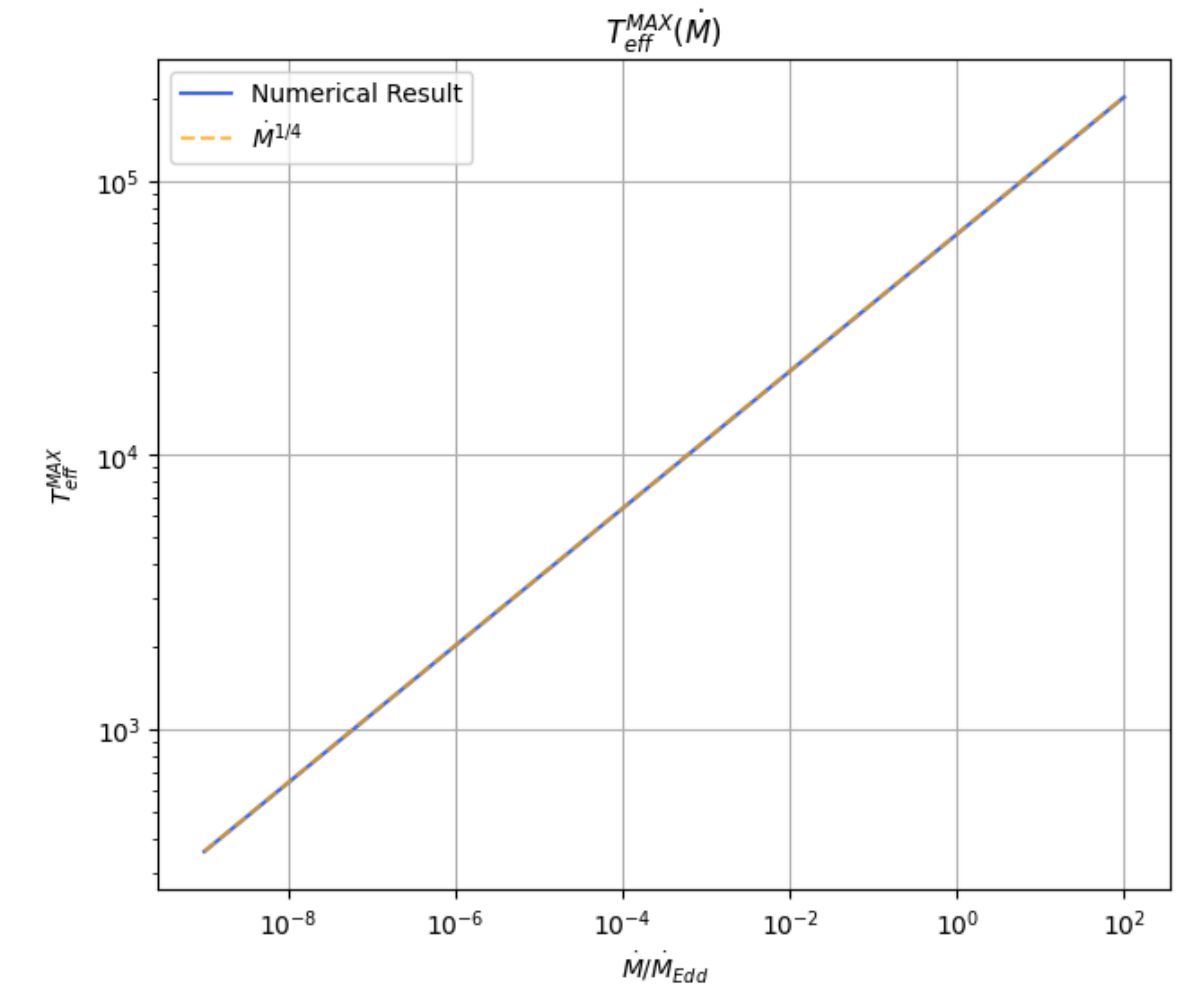
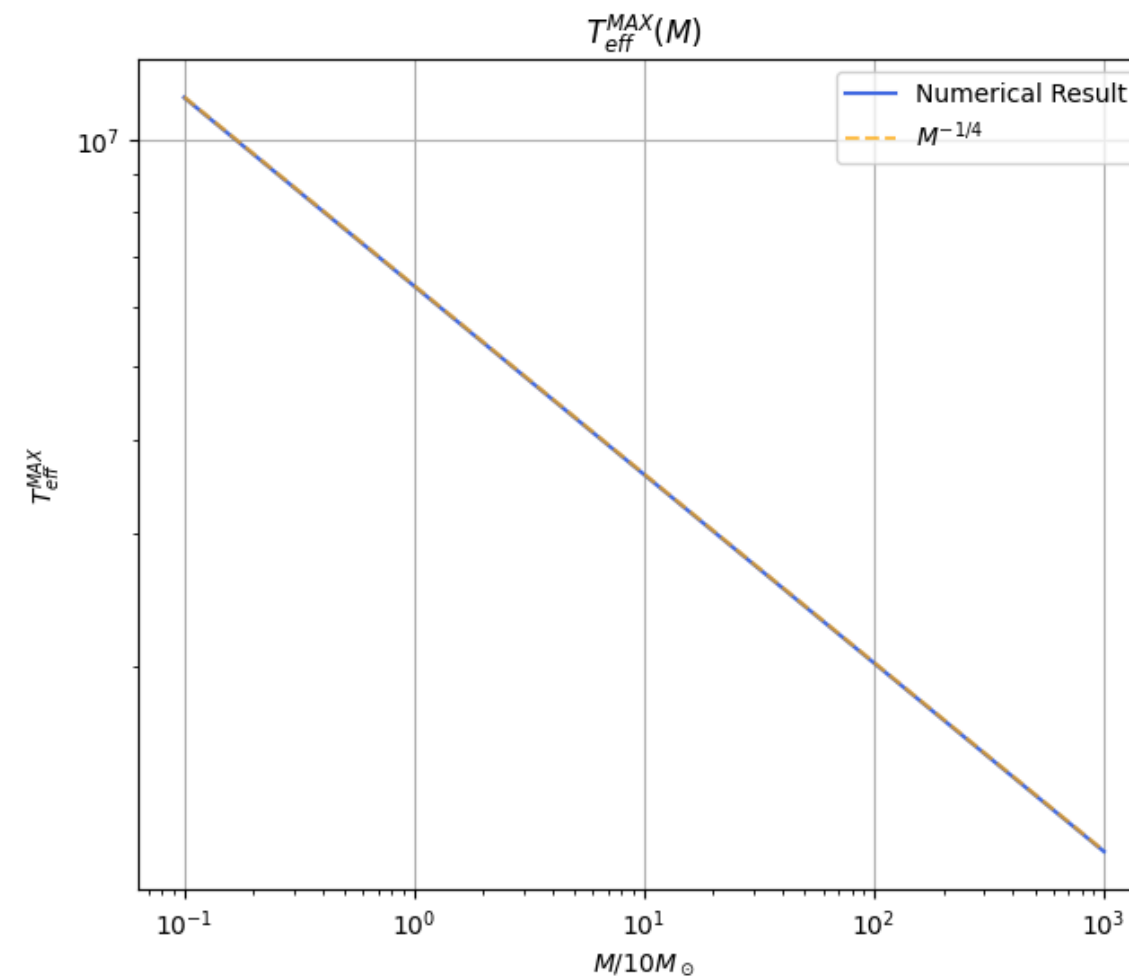
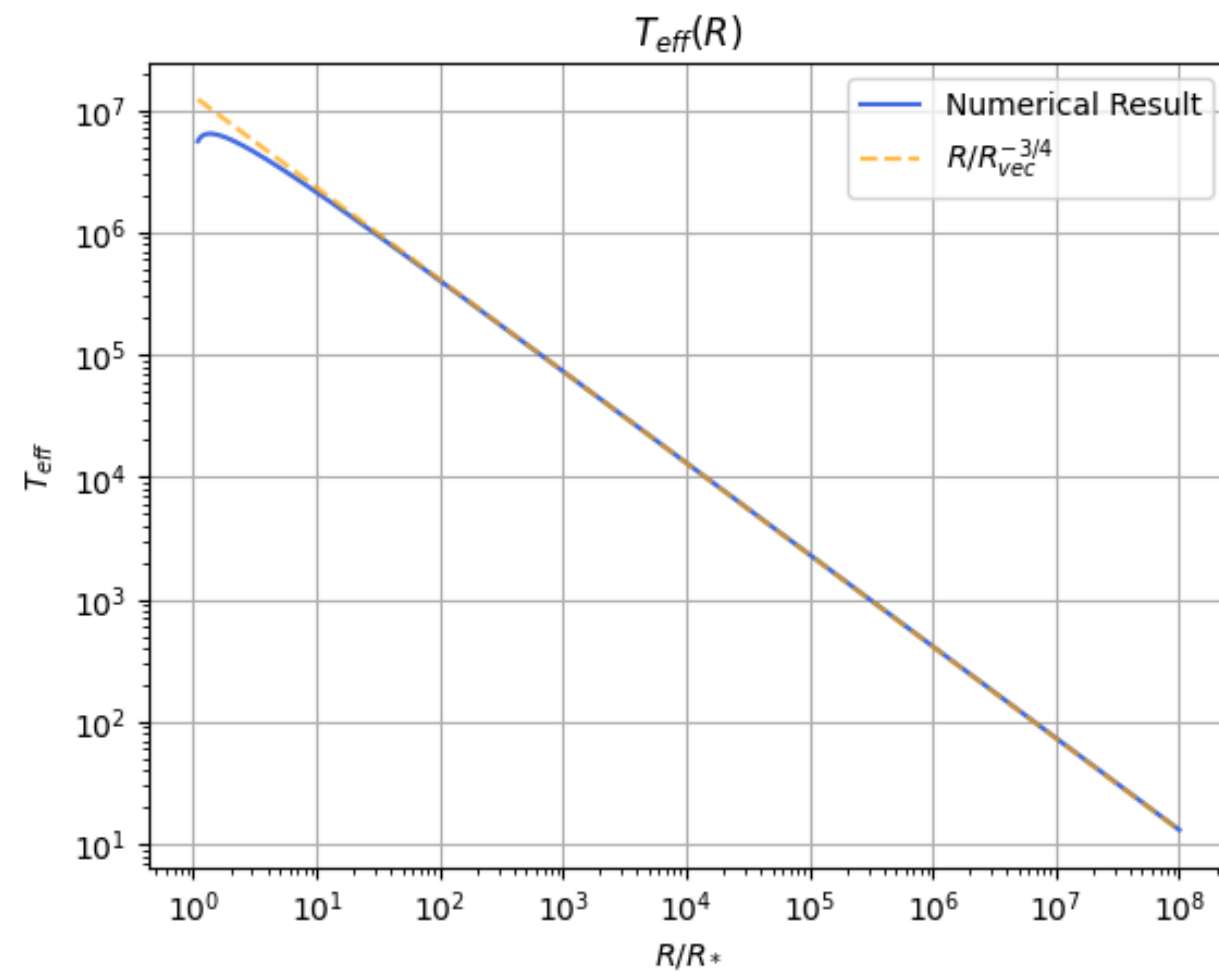
Peak Temperature

$$T_{eff} = \left(\frac{Q_+(R)}{2\sigma} \right)^{1/4}$$

$$\begin{cases} \alpha = 0.1 \\ M = 10M_{\odot} \\ \dot{M} = 0.1\dot{M}_{Edd} \end{cases}$$

$$\begin{cases} \alpha = 0.1 \\ \dot{M} = 0.1\dot{M}_{Edd} \end{cases}$$

$$\begin{cases} \alpha = 0.1 \\ M = 10M_{\odot} \end{cases}$$

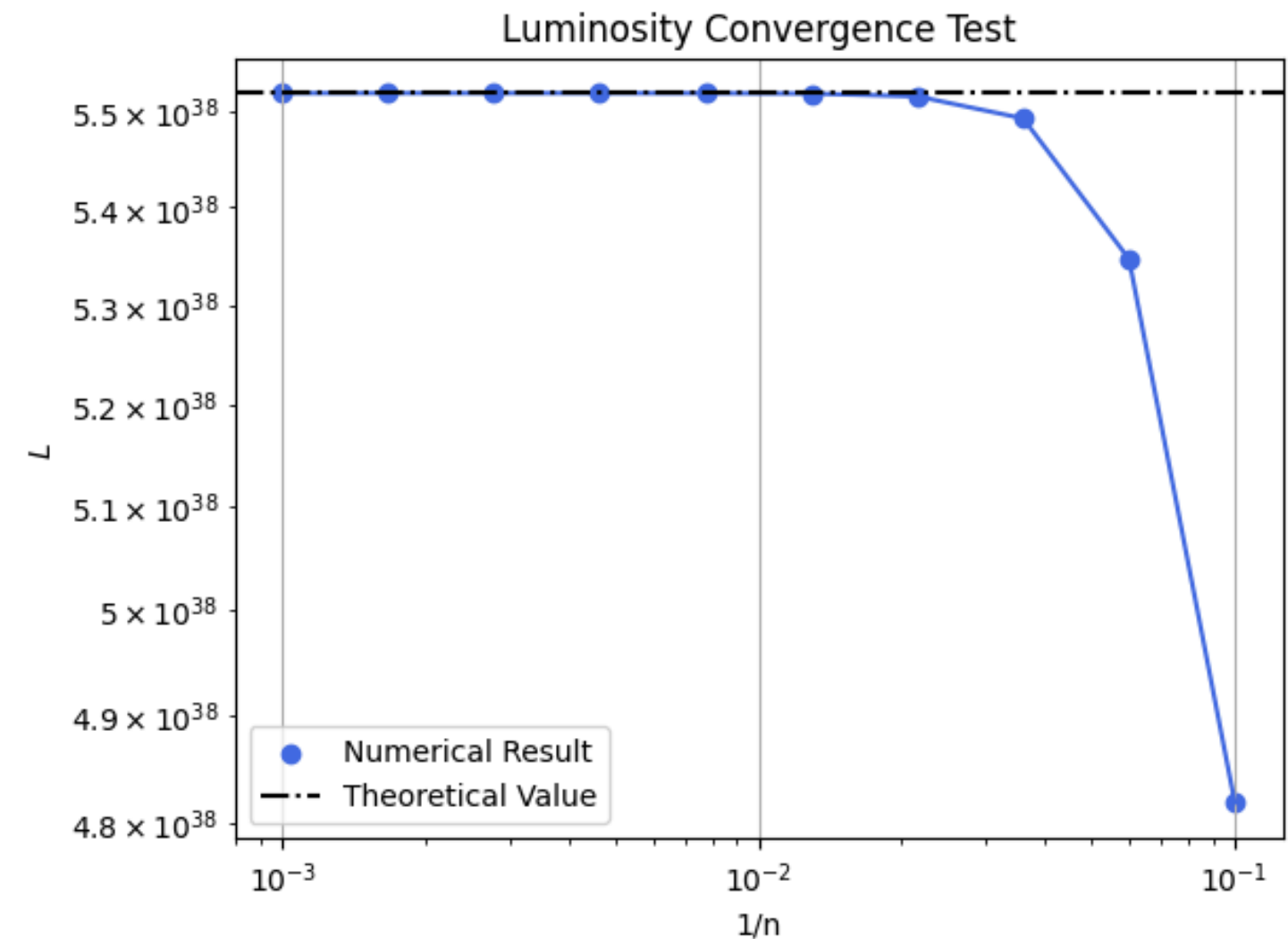


Luminosity Profile

$$\frac{dL}{d\ln R} = 2\pi R^2 Q^+(R) \rightarrow dL = 2\pi R Q^+(R) dR$$

I want to integrate the expression and check that the total luminosity agrees with the following analytical solution.

$$L = \frac{GM\dot{M}}{2R_*}$$



True Luminosity: 5.519043447751862e+38 erg/s

Numerical Luminosity: 5.519043266058506e+38 erg/s

Emergent Spectrum

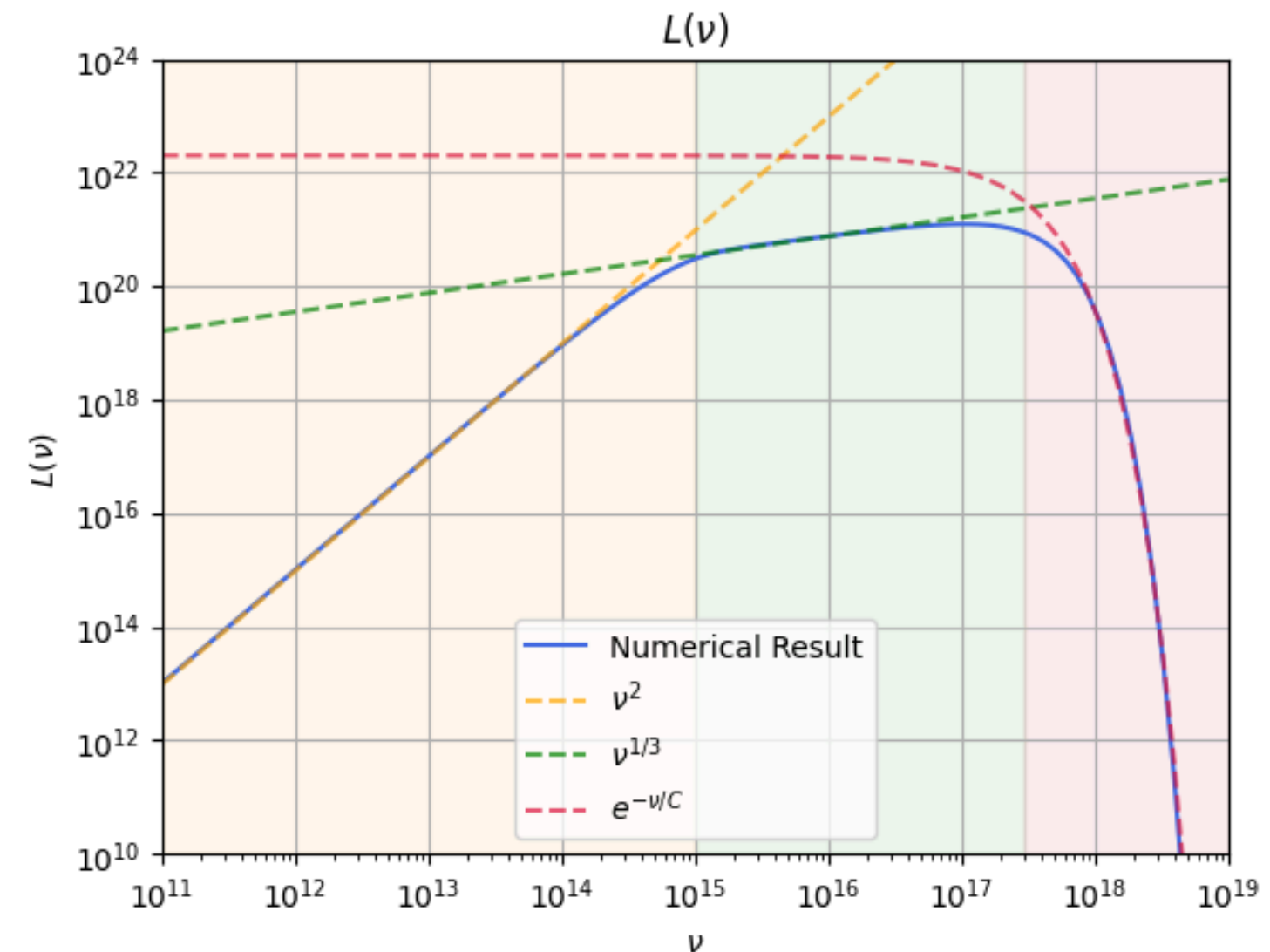
$$\begin{cases} \alpha = 0.1 \\ M = 10M_{\odot} \\ \dot{M} = 0.1\dot{M}_{Edd} \end{cases}$$

When looking at the emergent luminosity from a Shakura Sunyaev α accretion disk as a function of frequency, I expect to identify three regions:

1. Rayleigh-Jeans Tail: $L_{\nu} \propto \nu^2$
2. Multi-colour Disk Regime: $L_{\nu} \propto \nu^{1/3}$
3. Wien Cutoff: $L_{\nu} \propto e^{-\nu/C}$

$$B_{\nu} = \frac{2h\nu^3}{c^2} \frac{1}{e^{\frac{h\nu}{kT_{eff}}} - 1}$$

$$dL_{\nu} = \pi B_{\nu} 4\pi R dR$$



Conservation of j

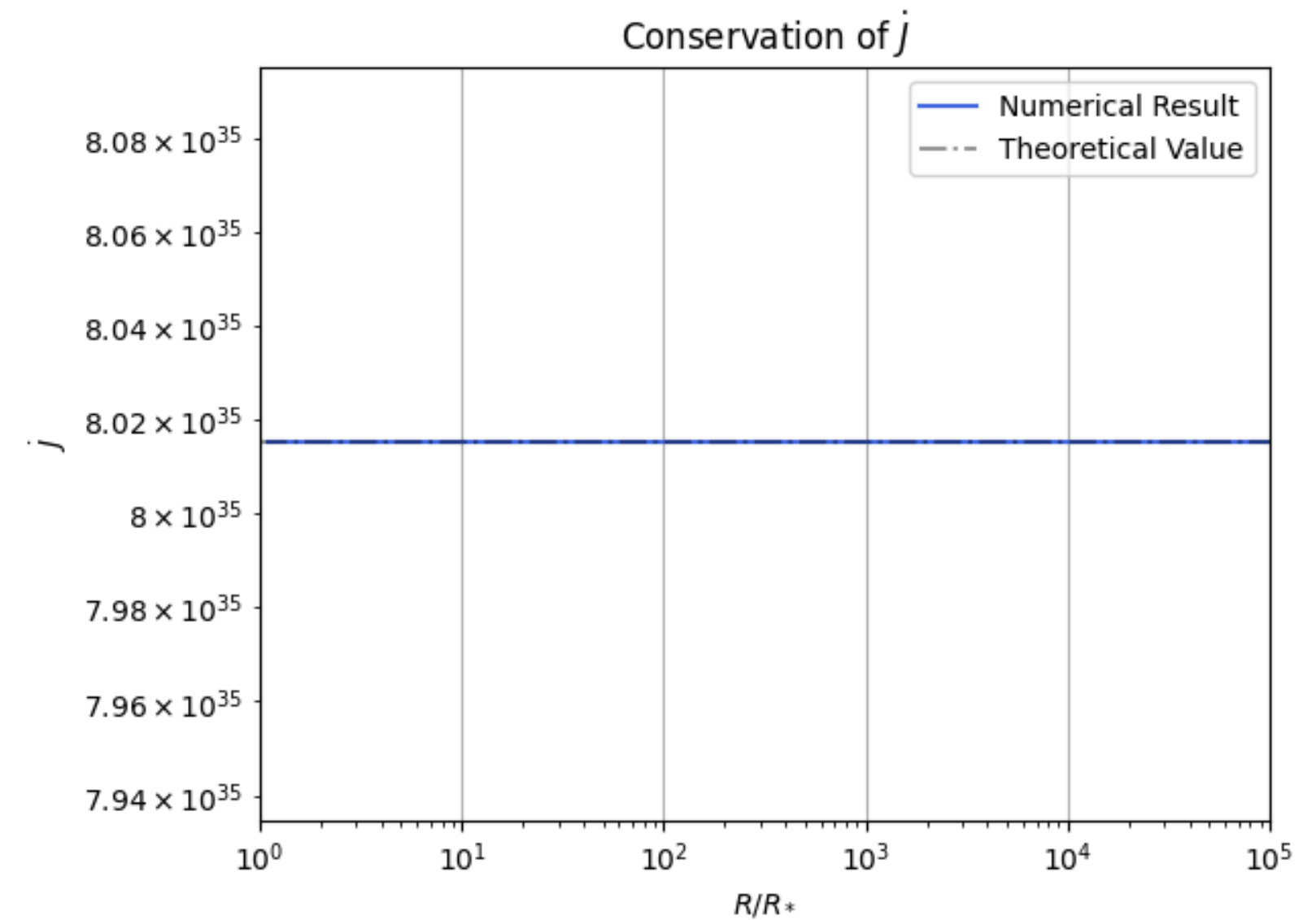
$$\begin{cases} \alpha = 0.1 \\ M = 10M_{\odot} \\ \dot{M} = 0.1\dot{M}_{Edd} \end{cases}$$

Looking at my system, I expect the flux of angular momentum to be conserved at all radii. I check for it to verify that my numerical disk is correct.

$$\dot{j} = \dot{M}\sqrt{GMR} - G_{\nu} = \dot{M}\sqrt{GMR} - \frac{8\pi R^2 Q^+}{6\Omega_K}$$

$$\dot{j} = \text{constant} = \dot{M}\sqrt{GMR_*}$$

As we can notice, the angular momentum flux is constant and matches the theoretical expectation.



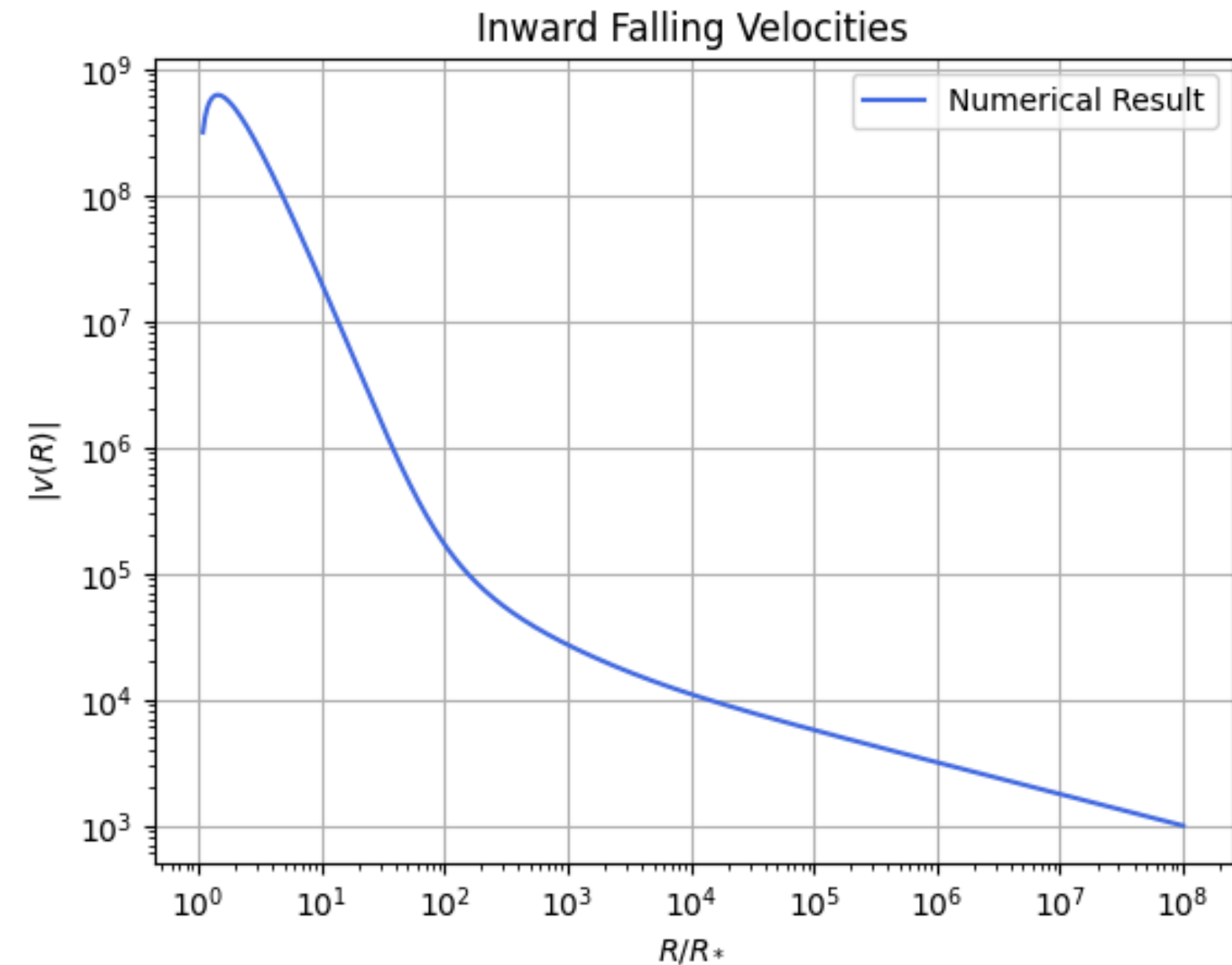
Infalling Velocities

$$\begin{cases} \alpha = 0.1 \\ M = 10M_{\odot} \\ \dot{M} = 0.1\dot{M}_{Edd} \end{cases}$$

$$\dot{M} = -2\pi R \Sigma v(R)$$

$$v(R) = -\frac{\dot{M}}{2\pi R \Sigma}$$

The inward falling velocities of the accreting material can get really fast, about $0.02c$, close to the black hole ISCO, but decrease rapidly outwards, where the disk is dominated by its rotation.

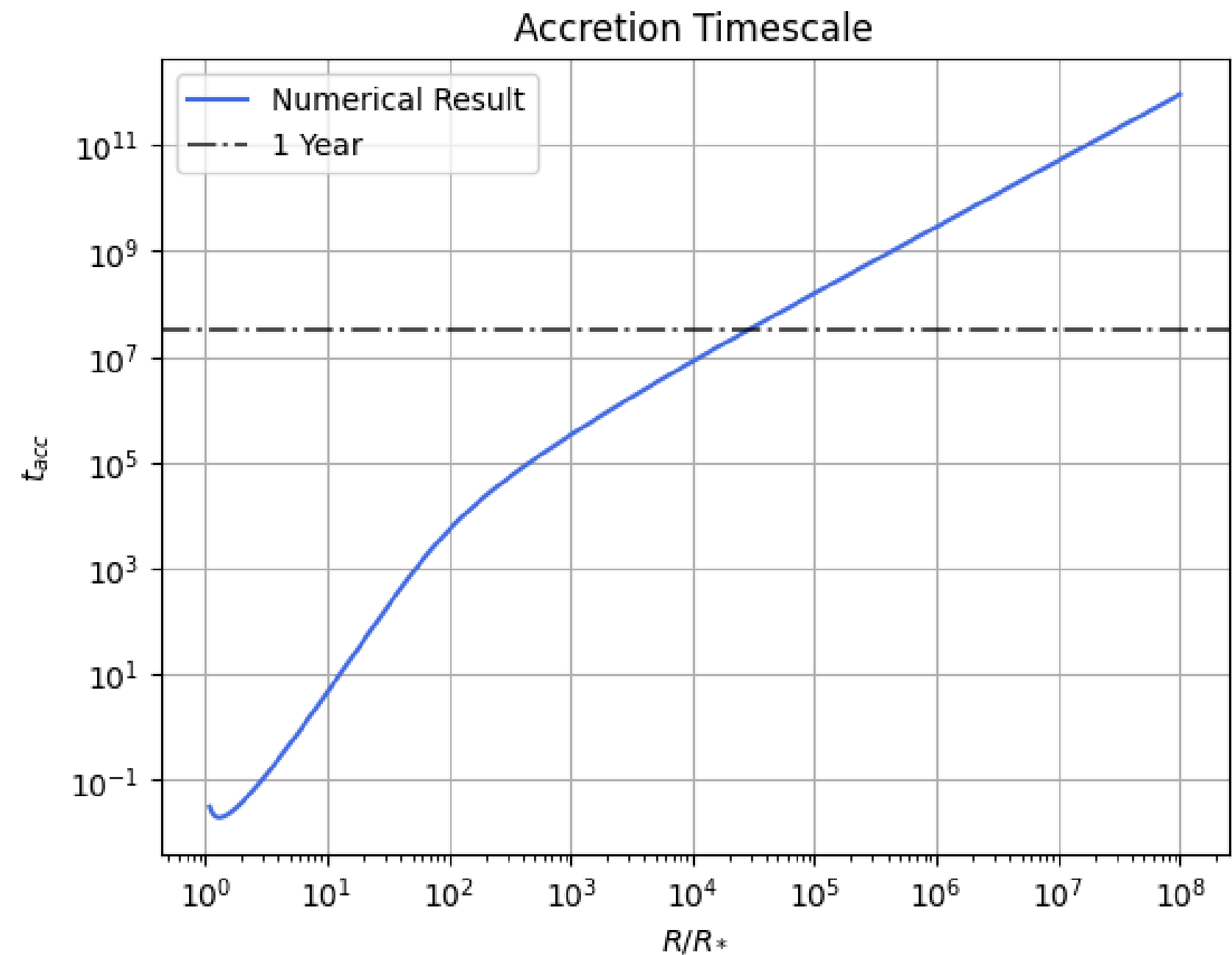


Accretion Timescale

$$\begin{cases} \alpha = 0.1 \\ M = 10M_{\odot} \\ \dot{M} = 0.1\dot{M}_{Edd} \end{cases}$$

$$t_{acc}(R) = \frac{R}{v(R)}$$

The accretion timescale increases, as one would expect, with radius. At a radius of about $2 \times 10^4 R_*$ the infalling mass takes 1 year to reach the black hole ISCO.



Bibliography

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**THANKS FOR THE
ATTENTION**